A Web-Based Gamified Fitness System Which Enhances Motivation and General Well-Being

Tejesh Ramesh Bawa
Student ID: 001173097-8

Supervisor: Christine Du Toit

BSc (Hons) Computing

Final Year Individual Project COMP1682

A dissertation submitted in partial fulfilment of the University of Greenwich Undergraduate Degree Programme

BSc (Hons) Computing

Word Count : 12601 Words

Abstract

Many people find it difficult and challenging to embark on a fitness journey, whether it is the initial participation or a continuous engagement due to weakening motivation. This report delves on how to tackle this weakening motivation through gamification elements. Gamification elements can play a pivotal role in raising and sustaining motivation to promote a healthier lifestyle. this study proposes a comprehensive gamified fitness system as a solution to overcome common motivational hurdles like responsibility and boredom to enhance user engagement through challenges and achievements.

Acknowledgements

I would like to thank all the amazing people that have been there for me throughout the entire project. I would have unable to complete this project without the support and guidance that I received from the following people.

A special thanks to Mrs Christine Du Toit, who is my personal tutor and supervisor. I am extremely grateful for her invaluable guidance, support and insight feedback throughout the duration of this project. Their expertise and encouragement helped to shape direction to this project.

I'd like to thank Mrs Keeran Jamil for all the support that she provided for all computing students throughout these years. Her way of constant remainders and lectures truly help understand what was required for this final year project.

I would also like to thank the University of Greenwich for providing the facilities, resources and environment for conducting my research. This atmosphere allowed me to gain invaluable learning experience and skills development.

I would like to than my family for the constant support, understanding and patience throughout this journey. They have constant supported me during my lowest moments and truly helped motivating me.

Finally, the project would not be completed without my circle of friends who constant push me to be the best .

Table of Contents

1. Introduction 6
1.1 Background 6
1.2 Aims and Objectives 6
1.2.1 Aims 6
1.2.2 Objectives 7
1.3 Approach 7
1.4 Justification of the Suitability of a Methodology or a Framework 7
2. Literature Review 7
2.1 Introduction 7
2.2 Gamification Effects on Motivation 7
2.3 User Interface Design in Fitness Application 8
2.4 Gamification Affecting System and UI Design 8
2.5 Body Positivity and Motivation in Fitness 9
2.6 comparative Analysis 9
2.7 Conclusion 9
3. Product Research 9
3.1 Introduction 9
3.2 Fitbit 10
3.2.1 Nielsen's Heuristic Usability Evaluation 10
3.3 Zombies, Run! (ZRK: Zombies Run + Marvel Move) 14
3.3.1 Nielsen's Heuristic Usability Evaluation 15
3.4 Jefit 20
3.4.1 Nielsen's Heuristic Usability Evaluation 20
3.5 Conclusion 25
4. Requirement Analysis 25
4.1 Introduction 25
4.2 High-Level Requirements 25
4.3 MoSCoW Prioritising High-Level Requirements 26
4. 4 Determining and MoSCoW Prioritizing the Functional Requirements 27
4.5 Determining the Non-Functional Requirements 28
4.5.1 Reliability 28
4.5.2 Security 28
4.5.3 Scalability 28
4.6 Entity Relationship Diagram 28
4.7 Use Case Diagram 30

4.8 Activity Diagrams 30
4.8.1 Registration of User 30
4.8.2 Users Starting a Quest 31
4.8.3 User Adding Posts on the Community page 31
4.8.4 User Reading the Admin Description or PDF 31
4.8.5 Admin Uploading PDF 31
4.8.6 Admin Editing Profile 32
4.9 Rich Picture 32
5. Design Development 33
5.1 Introduction 33
5.2 User Interface Wireframes 33
5.2.1 User Application 33
5.2.2 Admin Application 35
5.3 Mid-Fidelity Prototype 36
5.4 Design Elements 37
5.4.1 Colour Scheme 37
5.4.2 Fonts 37
6. Technical Project Development 37
6.1 Introduction 37
6.2 Flutterflow 37
6.3 Supabase 38
6.4 Changes Made 38
6.5 Technical Problems Faced 38
7. Legal, Social, Ethical and Professional Issues 38
7.1 Legal 38
7.2 Social 39
7.3 Ethical 40
7.4 Professional 40
8. Product Testing 41
9. Implementation 45
9.1 User Application 45
9.2 Admin Application 49
9.3 Database 50
10. Evaluation 50
Visibility of System Status 50
Match Between System and the Real World 51

User Control and Freedom....51
Consistency and Standards....51
Error Prevention....51
Recognition Rather than Recall....51
Flexibility and Efficiency of Use....51
Aesthetic and Minimalistic Design....51
Helps Users Recognise, Diagnose, and Recover from Errors....52
Help and Documentation....52
11. Critical Appraisal....52
11.1 Process....52
11.2 Product....52
11.3 Personal....53
11.4 Future development....53
12. Conclusion....54
References....55
Appendix 1: Project Proposal....57
Appendix 2 Comparison of Wireframes, Mid-Fidelity and Final Product....62

1. Introduction

1.1 Background

There is a frequent occurrence in the initial eagerness to start working out, which is usually sparked by one's individual goals or New Year's Resolutions, which rapidly dims due to a lack of motivation. According to Webley (2012), one of the most commonly broken New Year Resolution is “Loss Weight and Get Fit” which is further elaborated by McGonigal (2011) that a large number of people would sign up to gyms in January, but that number would be diminished by mid-February.

There are many different motivational factors that affect people performing any physical activities. Capel et al. (2015) asserted that there are primarily two types of motivations for engaging in physical activity. The first group includes things like tradition in the family, competitive motivation, parental support, socialising, goal-setting, tracking, and athletic background that are related to initial sports participation. The second group of factors, which included satisfaction of sports engagement, a sense of wellbeing, self-competition, interacting with others, health, and an abiding appreciation of nature, led to continuous sports activity.

Both McGonigal (2011) and Düsenflugzeug (2022) describes that people who get into fitness have set unrealistic goal which often clashes with their intention to indulge in physical activity. People require a combination of encouragement and realism which gives them the best chance of reaching their goals. This shows that many people run into multiple difficulties while trying to start a regular fitness schedule.

Ozdamli & Milrich (2023) explains that gamification features help motivate individuals to work out and that it has become a trend in the fitness sector. It is further emphasized that these gamification features add fun and a competitive element to regular workout routines which inspire and motivate people to achieve their fitness goals.

Neupane et al. (2020) states that many activity trackers have been using gamification features. It is agreed by both Ozdamli & Milrich (2023)and Neupane et al. (2020) that gamification features significantly help in motivating people to improve their health.

A potential solution to this issue would be to develop a system that aims to increase and maintain motivation, get past common obstacles to motivation (such as responsibilities, monotony, and detachment from fitness goals), improve user engagement through challenges and achievements, encourage a healthier lifestyle, and promote knowledge of the role that gamification plays in encouraging people to exercise and, ultimately, improve their health and well-being.

Keywords: [Fitness, Motivation, User-Interface, Feedback, Gamification, Game, System-Development]`

1.2 Aims and Objectives

1.2.1 Aims

This project aims to develop a system that adopts gamification features to address the lack of motivation in individuals who often encounter when trying to engage in physical activities.

1.2.2 Objectives

The objectives are shown in Appendix 1: Project Proposal

1.3 Approach

The project will adopt the agile methodology which is more flexible and adaptable throughout the development process. Regular user feedback and iterative changes will also be possible with this technique. This methodology will guarantee that adding gamification elements to a fitness-based system will be done in a systematic, user-centred, and flexible manner.

1.4 Justification of the Suitability of a Methodology or a Framework

In order to build a flexible and dynamic framework, the project will use the Agile Rooted Methodology in conjunction with the iterative development approach. The agile methodology will allow frequent feedback and adjustable changes, which make the development more responsive to both user needs. During the development of this system, the developers will be using evolutionary prototyping, which will first start with an initial basic prototype, which will further evolve and expanded upon the user feedback and requirements.

The developers will utilize Test-Driven Development(TDD) to prioritize more on quality and reliability which will concentrate on the quality of the code. Furthermore, the development team will apply UML (Unified Modelling Language) to design the system structure to foster efficient and effective development.

The development team will follow an iterative development process. This method allows for more flexibility and adaptation which associates the system closely with changing user requirements. By following this comprehensive approach, the system seeks to provide a high-quality and responsive fitness system to address the user needs effectively.

2. Literature Review

2.1 Introduction

The modern lifestyle facilitates an increase in sedentary habits and reduce regular physical activities. Speakman (2020) explains that modern smart technology may be the solution to address to the lack of motivation in maintaining regular physical activities. Smart devices and digital technologies have made it possible to develop a wide range of fitness application however fading motivation is still an issue as it stands as a significant obstacle for a long-term participation, this literature review will investigate on how to overcome lack of motivation through the relationship of gamification and UI design.

2.2 Gamification Effects on Motivation

Video games have become an important part in our daily lives explains Sebastian et al. (2021). It is further explained that game elements and mechanics can be used to improve user motivation and engagement in non-game environments such as fitness. Gamification can be used to leverage common game like elements such as competition, rewards, challenges and social interactions to transform traditional activities by tapping into intrinsic motivational factors and encourage a sense of achievement, enjoyable progress, and rewarding experience to users.

Rajšp et al., (2020) supports Sebastian et al. (2021) claims that gamification is a tool which increases user motivation and engagement. There are several types of gamification elements such as badges,

points system, leaderboard, rewards and progress tracking to name a few. However, the effectiveness of these element widely depends on the user's preference describes Razali et al. (2020). Agreeing with Razali et al. (2020) claims, Feng et al. (2020) adds that for game elements effectiveness can be categorized into two types of game elements commensurate and incommensurate. Furthermore, this study depicts that incommensurate game elements increase the user's intrinsic motivation.

Engaging in activities that are innately thrilling or enjoyable is known as intrinsic motivation says Razali et al. (2020) to determine which gamification elements to use, it is essential to discover the intrinsic motivation factors which could foster both the user engagement and loyalty towards fitness adds Feng et al. (2020). Razali et al. (2020) further explains that intrinsic motivation is important because it encourages users to act in certain ways that would benefit themselves.

2.3 User Interface Design in Fitness Application

User Interface (UI) Design plays a vital role in moulding user experience and satisfaction. A study conducted by Wang et al. (2019) provides interesting views on various components of UI and interaction design and how it can potentially influence user satisfaction and motivation habits. A supporting claim put forward by Zhou et al. (2018), even though UI design is crucial, personalization functionalities are just as significant for user engagement. The research emphasizes on the importance of accommodating individual's user preferences in functionalities such as fitness goals, workouts and self-monitoring.

Wang et al. (2019) study investigates in specific interaction and UI design approach and improve user experience in three distinct stages (initial trial stage, habit-forming stage and habit keeping stage). Huang (2022) perspective expands on this by suggesting that data visualization can function as both analytical and persuasive to users. The study emphasizes in the Motivational Design Factors (MDFs) which are motivational factors that can impact the user's motivation regarding fitness.

The findings of Wang et al. (2019), Zhou et al. (2018) and Huang (2022) becomes very crucial when understanding user motivation, satisfaction and engagement. By combing these different concepts together, it can create a comprehensive user experience which can align with individual user preference and UI design elements to contribute to a pleasant rewarding fitness experience.

2.4 Gamification Affecting System and UI Design

When adding gamification elements into a system and its UI design it is important to check and address the primary problem of the system and understand how suitable gamification elements are with the system says Wei (2017). Looyestyn et al. (2017) adds in that gamification elements combined with a well thought out user inface can affect user engagement.

Wei (2017) explains that before a system is integrated with gamification, a detailed requirement analysis is significant in the initial stages which can help to determine to choose the correct functionalities for the system that requires gamification. Furthermore, Looyestyn et al. (2017) study explains and delves deeply into the efficiency of gamified system and improving engagement by integrating constructive interaction between UI design and gamification elements.

An alternate perspective by Yamani (2021) challenges the idea of gamification universally strengthening user engagement. This research delves into the gamification framework and explains the positive features of gamification. However, it emphasizes the fact that it is not practical to make everything into a competition and force people into situations where they have to collect points for change. It is important to take a systematic approach to gamification when developing the system and UI design to create an environment where people can actively participate and enjoy.

When comparing the findings of all three studies, there are many similarities between them which helps to understand how gamification can affect the system or UI. These gamification elements can resonate differently with individuals who have specific design choices even though a well-designed UI is developed which can influence the overall structure of the system.

2.5 Body Positivity and Motivation in Fitness

As people get more concerned with their physical appearance, committing to promote body positivity is necessary for many fitness applications. According to (Sun, Jiang and Huang, 2023) body satisfaction is subjective and depends on one own appraisal of body image that fulfils their own physical attractiveness. The research concludes with describing the use of fitness apps to relate to body satisfaction positively.

Leung et al. (2023) provides valuable insights on how body image flexibility would positively link to exercise behaviour which is fuelled by autonomous motivation (personal value) or negatively linked to controlled motivation (internal pressure). The research also emphasizes to consider the extent of psychological experience of individuals can be different.

When integrating the system, it is to seamlessly meld body positivity together with motivation and behaviour. Both researchers help understand the significance of diverse body types and account for the user's psychological experiences. This will help individuals to be motivated as well as understand the importance of self-acceptance, self-worth and mental well-being.

2.6 comparative Analysis

After exploring several studies in the previous sections, it reveals insights and helps understand the interaction in between gamification, UI design and body positivity within fitness apps. A common theme has emerged which highlights the importance of gamification elements, and how positively it can impact users' engagement and intuition. The synthesis of gamification and UI design can function as a blueprint to understand user interaction and motivation to go beyond general principles.

Across these multitude of studies, an emphasis to gamified elements and social behaviours acting as potent motivators. The studies, also mentions the impact of UI design and the need of tailored approach which aligns with each individual user. These help to recognize the trend and allow to make well informed decisions when developing product in order to cover gaps which have been missed by previously created fitness apps.

2.7 Conclusion

In conclusion, the literature review explores gamification, user interface (UI) design and body positivity within fitness apps. By leveraging these findings, the developed fitness app can help create an environment for users to not only motivate but also build a sense of belonging and empowerment. All this will promote fitness engagement through the use of gamification elements.

3. Product Research

3.1 Introduction

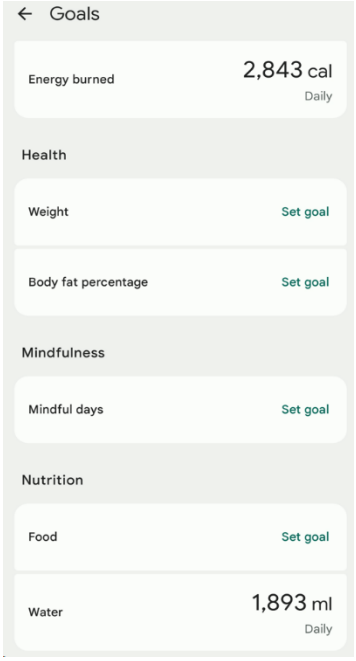
In this section, an analysis is conducted of different mobile fitness application within the area of study. The apps that are selected for this section will accommodate a variety of popular applications that are able to provide a diverse range of workout routine accompanied with gamification elements.

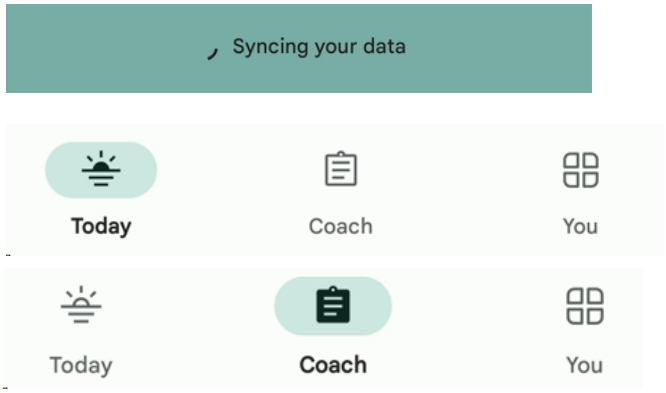
The criteria used is the Nielsen's Heuristics Usability Evaluation that will help to understand and improve user experience as well as the overall quality of the developed product.

3.2 Fitbit

Fitbit is one of the most popular fitness tracking applications that helps people monitor and boost their physical activity , sleep and overall health. Fitbit's approach towards health and fitness is done by integrating the system with different gamified elements that in turn motivates their users by turning regular, boring and traditional fitness activities into enjoyable experiences.

Fitbit makes use of a points system, where users can earn points by completing different types of workout routines. Users are able to enter ranking and leaderboards by these points allowing users to





compare their progress with other people to create a sense of competition. Another gamification strategy incorporated by Fitbit are challenges where users can compete with each other.

The app also utilizes virtual badges and achievements. The users can achieve them by reaching specific milestones or by simply completing challenges. These can act as a visual representation of the accomplishments and rewards whereby encouraging them to have a healthier lifestyle. Fitbit successfully goes beyond traditional fitness tracking application by adding the correct gamification elements to support and foster a healthy lifestyle.

3.2.1 Nielsen's Heuristic Usability Evaluation

Visibility of System Status

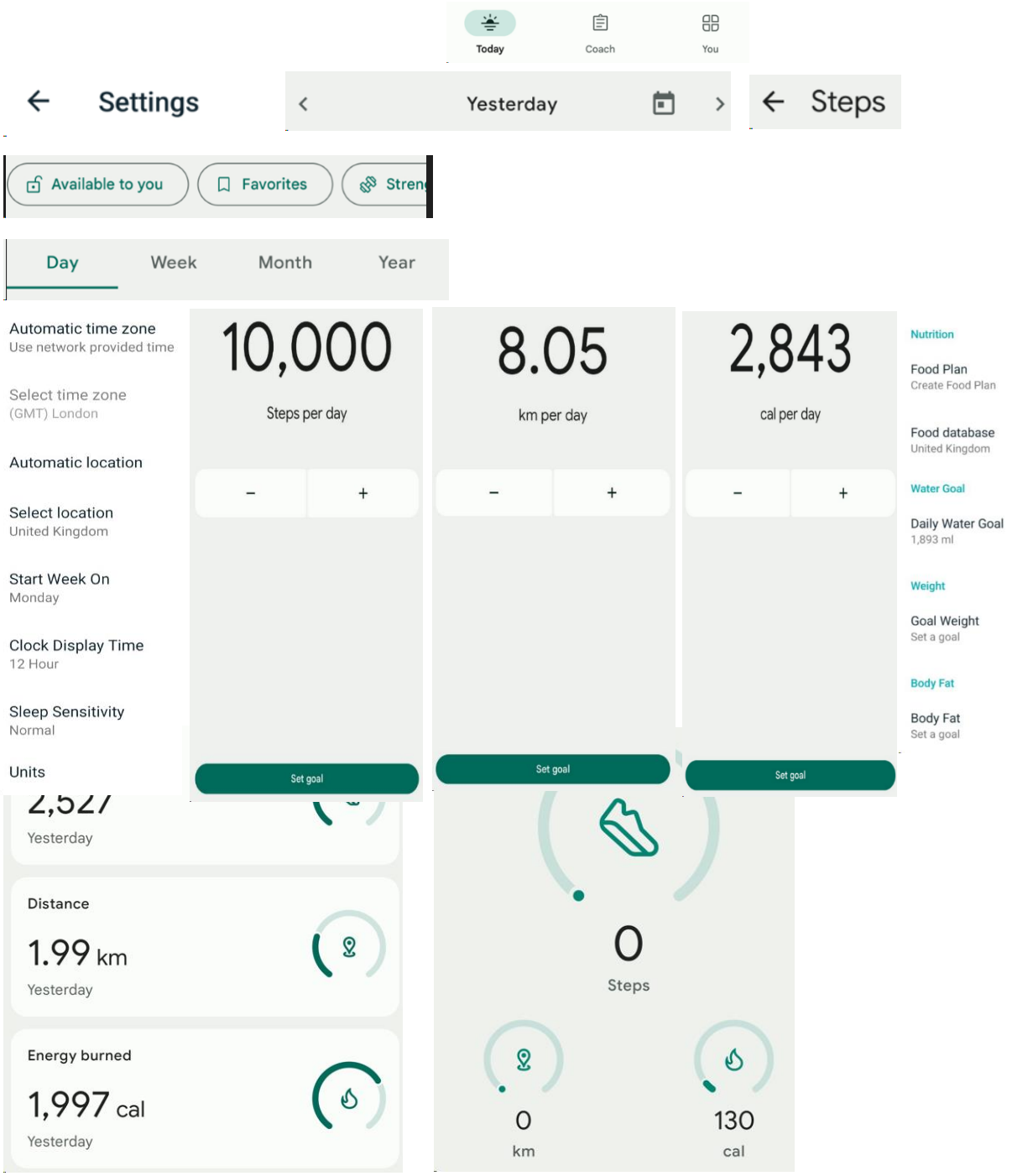
The application clearly communicates the visibility of the system status. The system automatically updates the step count, however when manually done, it launches a loading spinner with captions to indicate the progress of the system. When navigating to different pages, the app clearly indicates where the user is by the title of the page as well as indicating on the navigation bar at the bottom of the page highlights the icon to that specific page.

Match Between System and the Real World

The design uses words, terms, icons and images as which are familiar to the users. Many icons such as the steps, food, water and many others are common everyday objects as well as terms used in different pages are regular terminology used in fitness.

User Control And Freedom

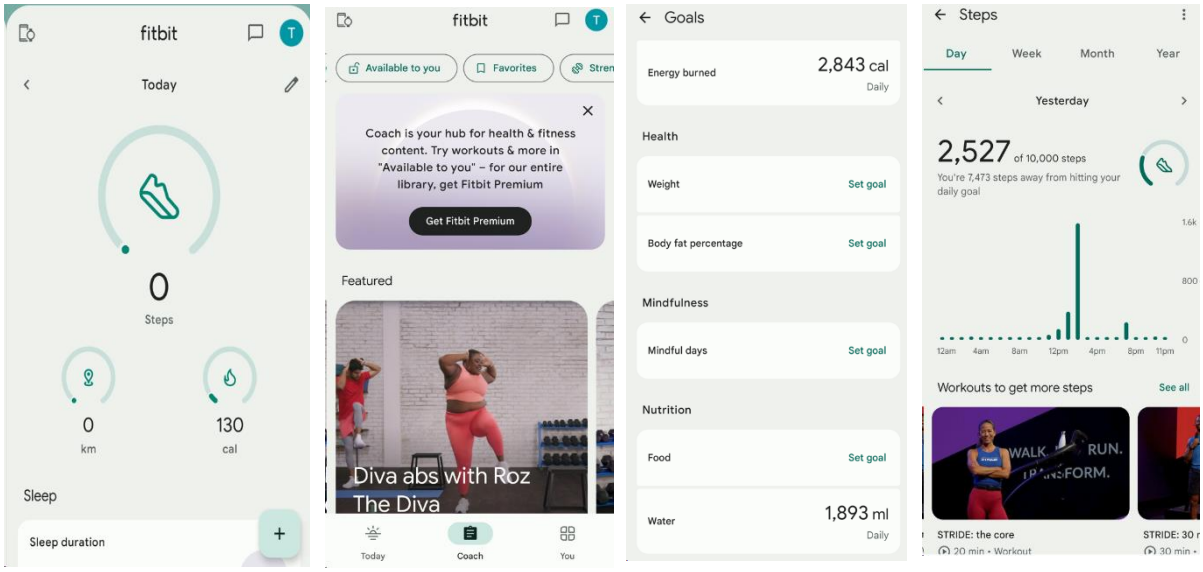
The application gives the user the capability to freely navigate throughout the system. Fitbit does this by letting their users to customize their own goals, preferences and metrics to foster a sense of freedom and confidence. The app makes use of back buttons as well as navigation panels as different parts of the page, which is easy to find, however to many of them can confuse the user. Furthermore,



the interface makes it easy for the user to navigate between features, pages and settings with ease and simplicity.

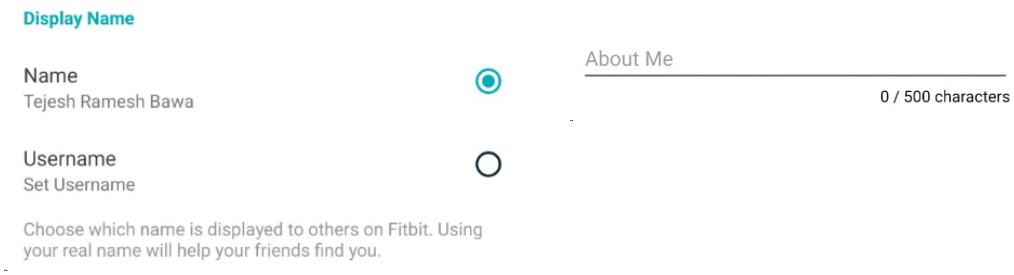
Consistency and Standards

The application maintains consistency by encouraging intuitive actions that reduces the cognitive load. Consistency is applied on the widget patterns, colour schemes and icons on different pages. As the pedometer is the main focus of the app and is similar to many other traditional pedometers. This standard makes the app and its functionalities very easy to learn as new users.



Error Prevention

Error messages are incorporated into the Fitbit design to prevent problems from arising in the first place. These can get very useful for new users as it helps prevent major changes to their preferences such as personal details, fitness log, etc. as it sends feedback and suggestions to users to take the correct approach. Fitbit implements logical explanations so that the user can easily understand what is required by them or show users how much to write or which options to choose.



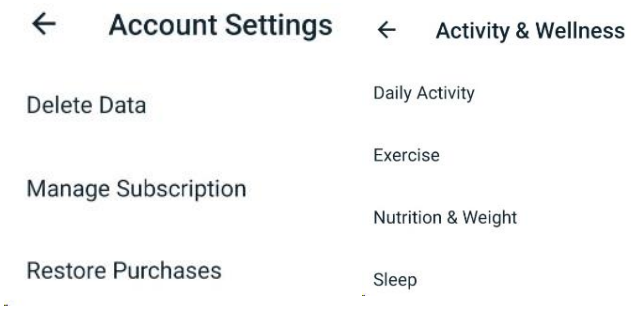
Recognition Rather than Recall

Fitbit successfully minimizes the memory load by making actions and information recognizable and intuitive rather than relying on memory. The dashboard summarizes the user's daily activities as well as their past activity log by displaying metrics like calories burnt, steps, calory intake, and water intake. This allows the user to quickly recognize their progress without the need to recall any of it.



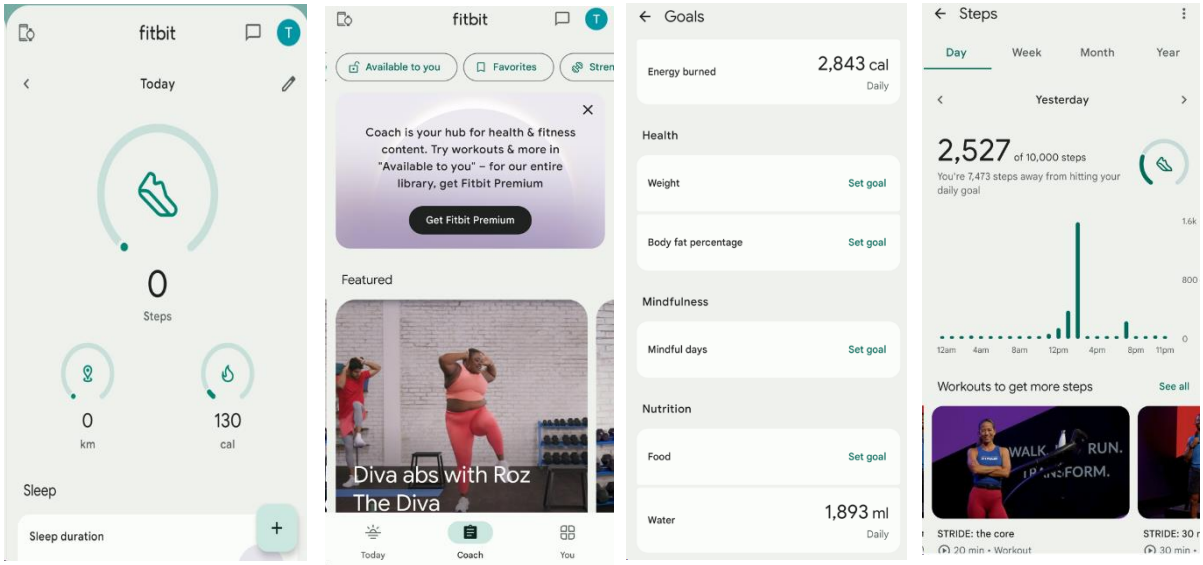
Flexibility and Efficiency of Use

Fitbit accommodates to different user preferences that provides a flexible user experience. This is observed when casual users can operate the simple health tracking metric whereas intense fitness users can access more advanced and detailed analysis. It is quick and easy to access the functionalities such as goal setting and daily activity summaries which improves the overall efficiency of the system. However, in many pages it is hard to distinguish between words and buttons, bringing the efficiency of the system down. Users may find it frustrating as they cannot visually differentiate between them which could easily have been solved by add borders.



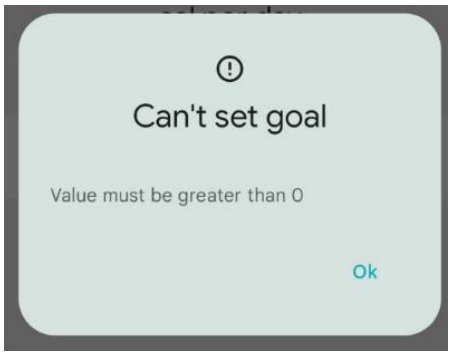
Aesthetic and Minimalist Design

The interface design developed by Fitbit is clean and aesthetically pleasing that doesn't divert the user's attention away from the main goal of the design. It makes use of a simple colour scheme, fonts, and layout which grants the system a minimalistic and clean design. The colour scheme compliments the aesthetic that Fitbit has created, which makes the user feel calm and relaxed.



Help User Recognize, Diagnose, and Recover from Errors

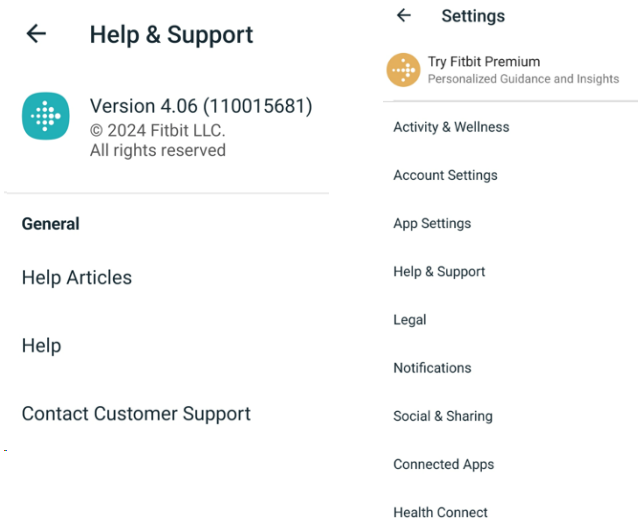
Fitbit's design and user interface provides users with informative error messages that addresses to any potential error made by the user. These messages can range from different problems such as connection problems, syncing errors, or obvious problems like setting a goal of 0.





Help and Documentation

The Fitbit application can get complex and confusing, in order to tackle that issue, Fitbit provides new users with adequate tutorials and guides to understand how the features and functionalities can be operated. Furthermore, the app is integrated with basic documentation and Frequently Asked Questions (FAQs) for users to access. The system also has an online community, where users can share different tips and insights.



3.3 Zombies, Run! (ZRK: Zombies Run + Marvel Move)

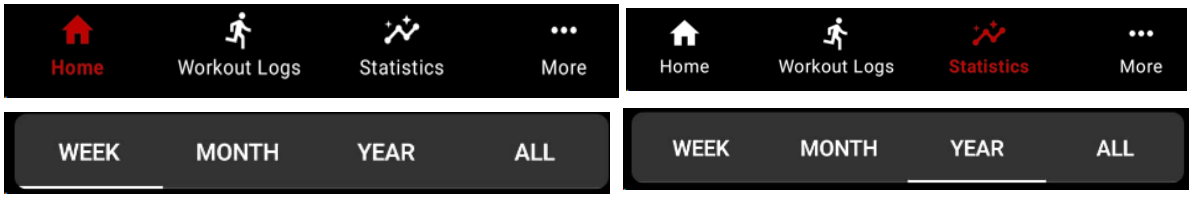
Zombies, Run! Is a well-known trilling running application that makes jogging more entertaining by infusing gaming elements together with a compelling storytelling. When the users run in the real world, they would listen through their headphones to a narration of an adventure where they are being chased by zombies.

The app incorporates gamified components like zombie chases and pursuits, narration driven quests alongside gathering items available in the game. Throughout their runs, user have to gather a variety of different resources, which in turn can be used to improve and expand their base within the app. This gives users a feeling of advancement and accomplishment. The app has been further developed and added Marvel stories, which allows them to feel like their favourite superheroes.

3.3.1 Nielsen's Heuristic Usability Evaluation

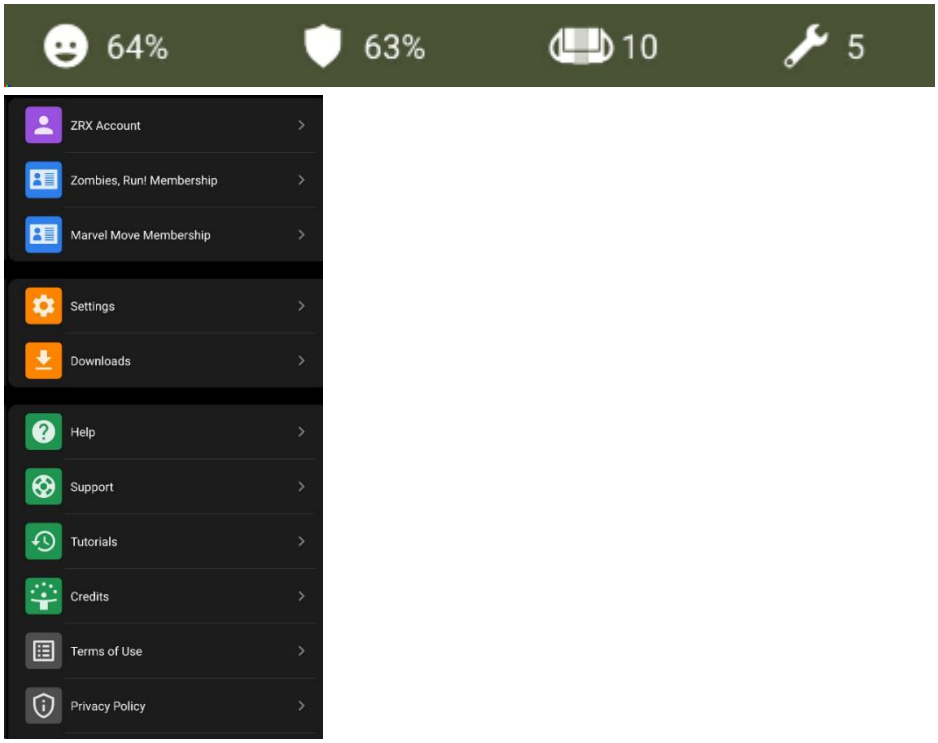
Visibility of System Status

The application adequately provides the user with real time feedback and system status regarding their activity progress. This is achieved through audible signals and visual indicators that conveys important information including distance travelled, time passed and their respective goals. However, the main dashboard does not show the fitness metrics and instead shows the plot of the story and the main gaming aspect of the system. The system also shows on which page the users is by highlighting that specific page.



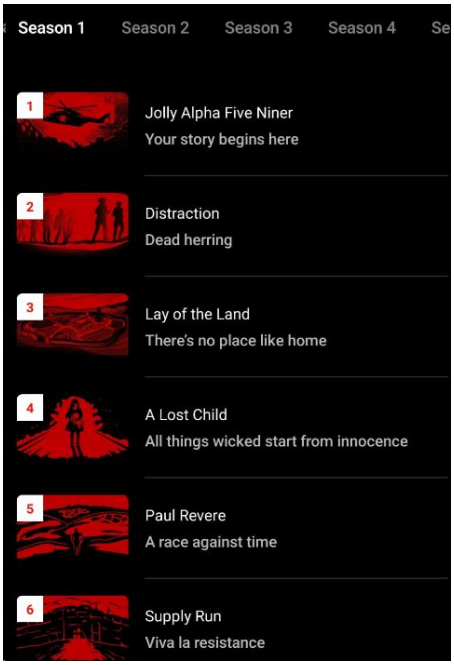
Match Between System and the Real World

The user interface design makes use of common everyday icons which the user can easily identify. The system successfully merges the virtual and physical world which align the fictional content to the real-world activities setting it apart from traditional fitness apps.



User Control And Freedom

The application strives to give users complete control and freedom by offering multiple types of customizable features like the ability to choose numerous training modes (different zombie episodes), personal goals and preferences (zombie chases). Also, the users have the ability to navigate throughout the entirety of the system without any difficulty. However, a major problem is the restriction users feel when adhering to the story line which may negatively affect their confidence in navigating the system. Users are able to choose a multitude of different story lines for their runs.



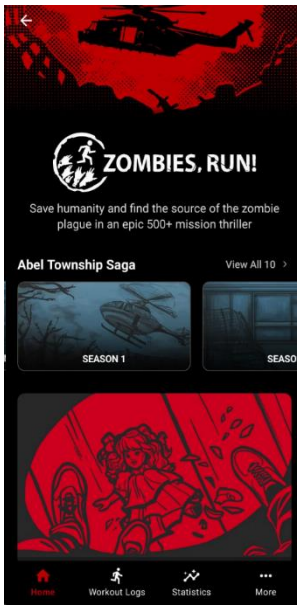


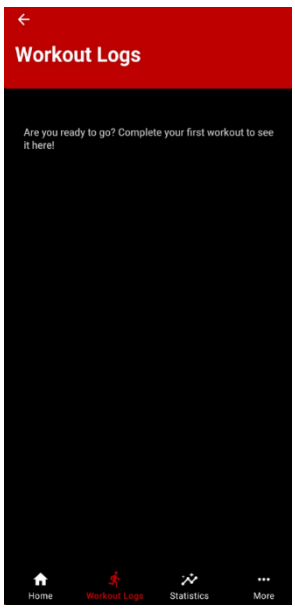


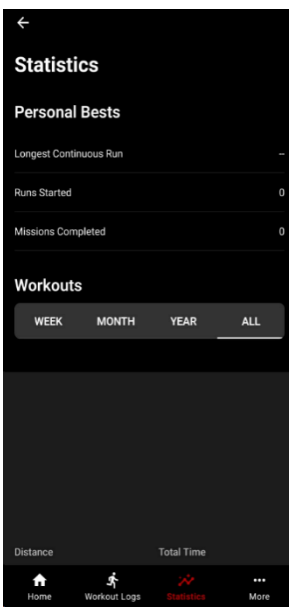


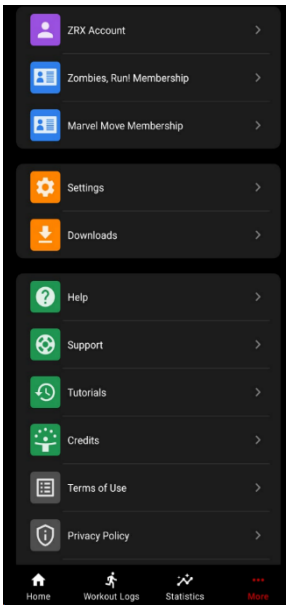
Consistency and Standards

The post-apocalyptic setting that is shown in Zombies, Run! plot reflects the design components which includes colour palettes, fonts, icons and themes. This strengthens the consistency by succeeding to engage user's narrative journey plus establishing an effective brand image. The design layout of the application remains similar throughout, from the home page to in-run presentations, making it easy for people to get around the system and immerse in the distinct setting it generates.





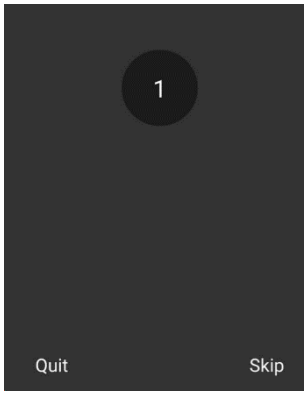


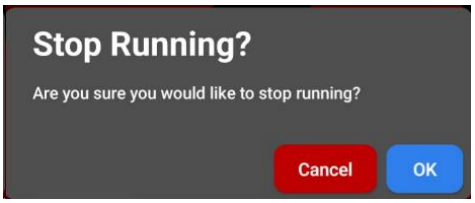


Error Prevention

To minimize the possibility of error during intense training sessions, the user interface is carefully designed to stop inadvertent taps and similar actions. Additionally, the applications strategically place these critical buttons to avoid missing parts of the story (by adding the replay option) and starting unwanted activities. This meticulous design focuses on details to bring a seamless and error free experience.

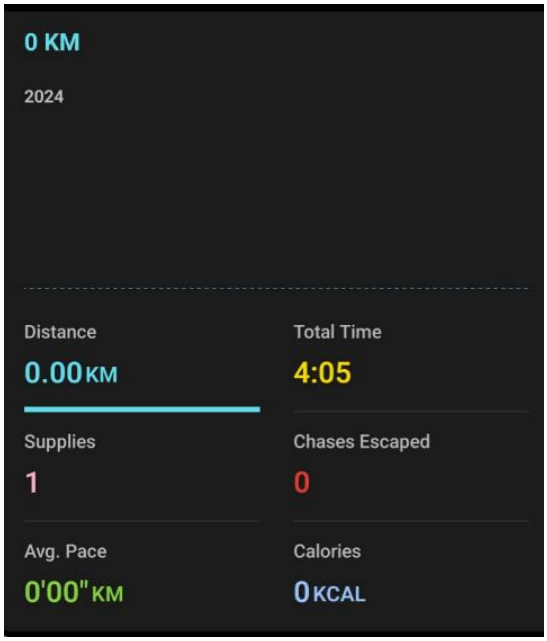






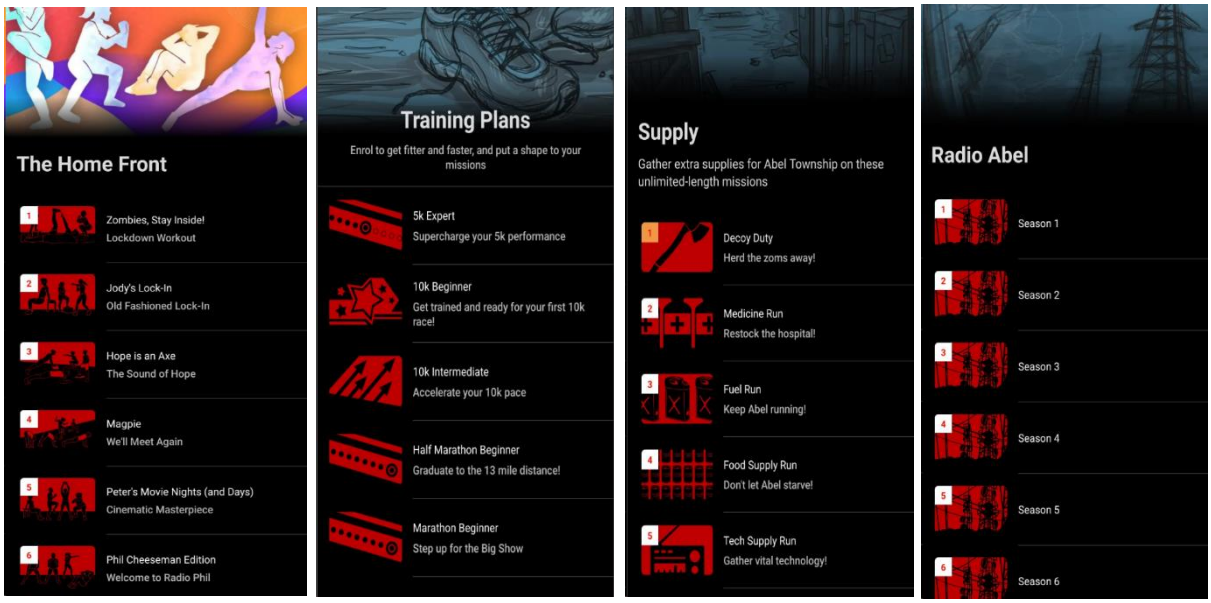
Recognition Rather than Recall

The app does not encourage its users to remember instructions or action and provides obvious indicators to walk users through its functionalities of the system. Buttons, icons and various other UI elements used are easily recognizable to allow users to quickly grasp various features instead of primarily relying on memory. It uses similar functionalities to Fitbit, which shows the distance, time taken, calories burnt and average pace. However, unlike Fitbit, this app does not utilise logos at that dashboard. By not doing this, it makes it harder for users to recognize what everything means and would have to rely on memories.



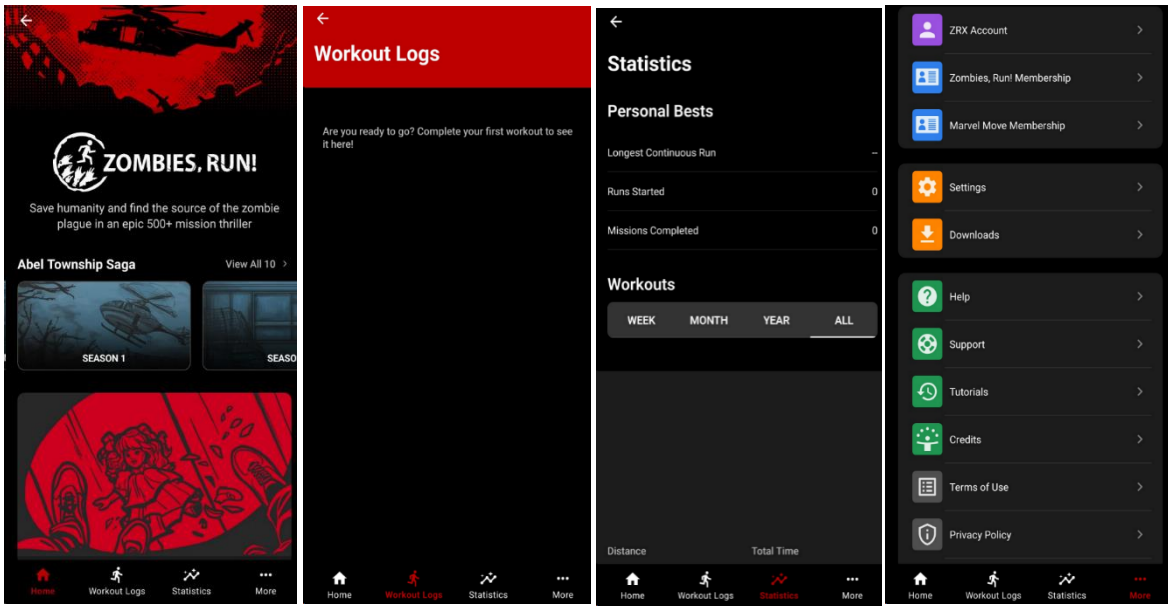
Flexibility and Efficiency of Use

The application provides adequate flexibility and efficiency to the users by giving them different types of workout options that the user can themselves tailor the exercises to meet their goals. The user-friendly interface guarantees people can effectively browse through various setting and stories allowing them to effortlessly change their styles and plots.



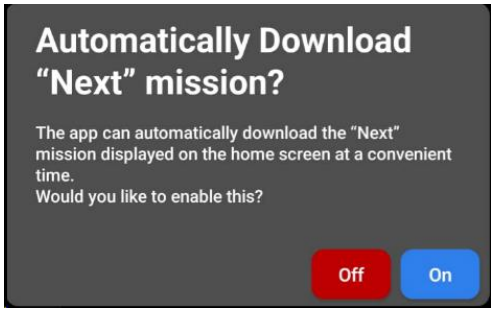
Aesthetic and Minimalist Design

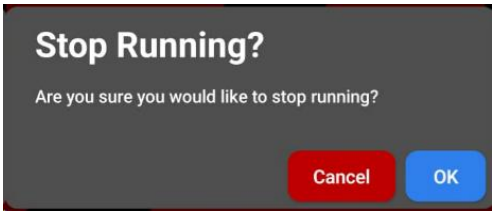
In keeping with its post-apocalyptic storyline, Zombies, Run! has integrated a simple, elegant style that further improves the user experience whilst rendering it aesthetically appealing. The application utilises a well selected colour scheme, font, and icons to produce a visually appealing and captivating environment. In the absence of confusing the user with extraneous information, the thematic design elements, such as zombie-themed graphics and background images. This overall improves the ambiance of the system. By simplifying user interface demonstrates the fundamentals of minimal aesthetic by avoiding needless cluttering and ensures that every piece has an independent function.



Help User Recognize, Diagnose, and Recover from Errors

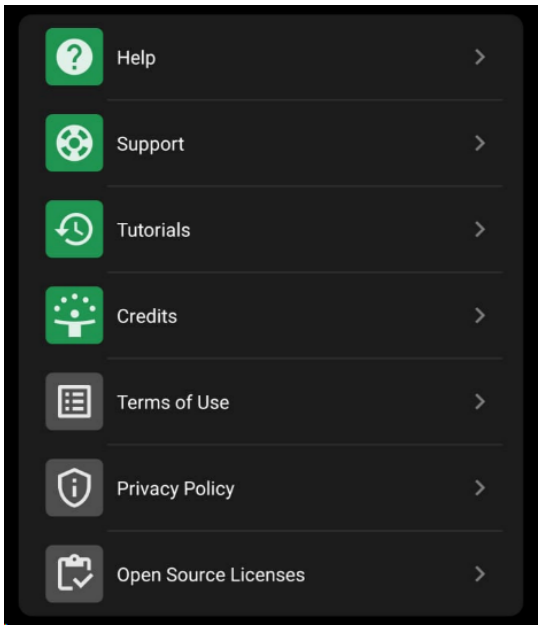
Zombies, Run! employs an interactive approach such as error pop-ups by including straightforward and educational feedback techniques into its design to assist users in identifying, diagnosing, and recovering from mistakes. These methods are placed in many different places where regular errors are made by user like the sign or log in page, personal details page or when there is a connection issue. When a problem occurs, the app gives users clear and concise error messages to help them figure out what's wrong fast. These are meant to be brief and simple enough for consumers to understand, assisting them in identifying the root of the issue rather than being confusing or frustrated.





Help and Documentation

The system incorporates an accessible help section in the profile page. Users may diagnose frequently occurring issues on their own and find solutions themselves with the help of a comprehensive FAQ area as well as guidance materials. Having a well-structured help collection increases user independence by enabling users to locate pertinent information whenever it's convenient for them. Zombies, Run! prioritises self-service options support and user education in order to ensure users are able to make full use of the software effectively and handle any questions or problems that might arise up along their fitness journey.



3.4 Jefit

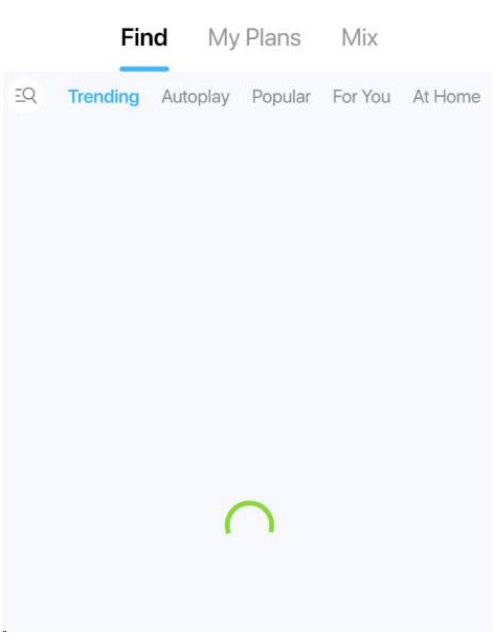
Jefit is a well-known fitness application that was created to assist users in tracking and maximising their exercise routines. the application offers different varieties of functions that help to accomplishing goals, which serves as an all-in-one tool that benefits novices and expert fitness lovers alike. In contrast to previously discussed apps primarily focus is on the pedometer, on the other hand Jefit lets users design and record their diverse exercise plans, by guaranteeing to monitor the performance and consistent behaviour. The app also has a vast workout database, which includes a variety of exercises that the users can use.

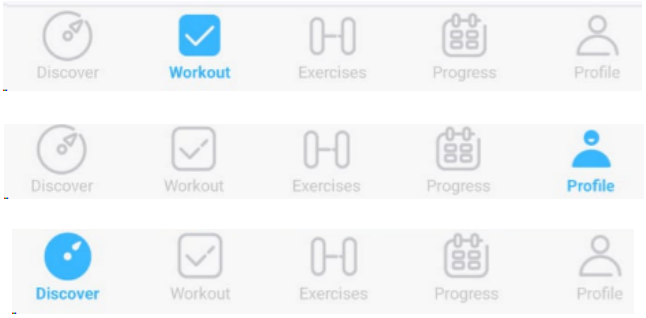
Jefit has incorporated gamification elements such as competition, point system and virtual badges. As previously mentioned in other apps, this help users to foster a sense of achievement. Furthermore, the application has integrated the social aspect of social media platforms like Instagram and twitter. This allows the users to post their achievements and progress to the online community which helps build a supportive network of people which can motivate and provide advice to each other.

3.4.1 Nielsen's Heuristic Usability Evaluation

Visibility of System Status

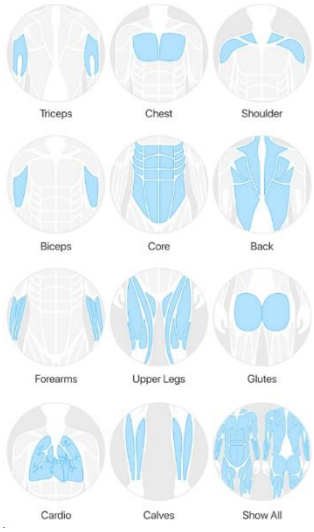
The application generally provides with satisfactory feedback for actions like when a workout is selected. When new pages are loaded or when the feeds in the discover page is refreshed there is a loading spinner indicating that process of loading process of the system. Furthermore, when transferring to different pages, the navigation bar at the bottom or the sub-tool bar in different pages indicate to the user by highlighting on which page they are on.





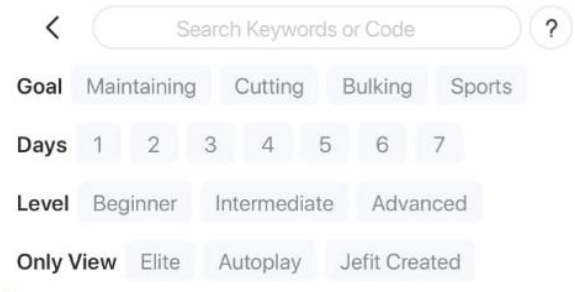
Match Between System and the Real World

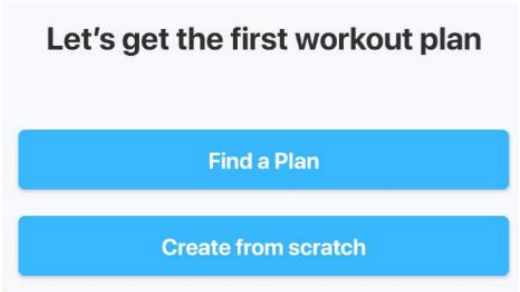
Jefit makes use of terminology and iconology that are well known in real world standards. When finding exercises to perform, the app provides images of the areas of muscles that the user might want to target. The app also gives clear visual and auditory instructions similar to fitness trainers at the gym. This enhances the user's understanding on how to operate this application



User Control And Freedom

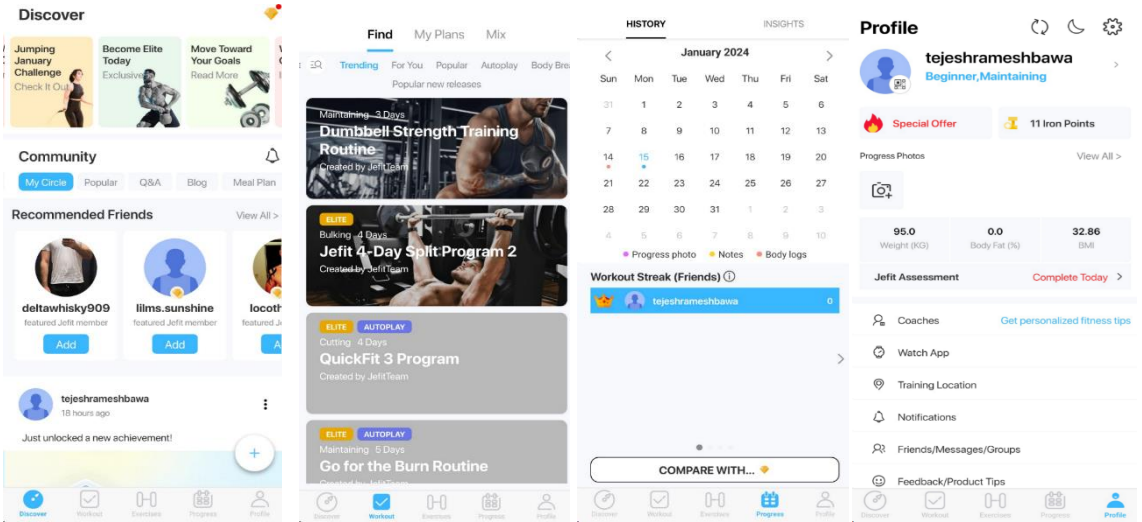
Jefit allows its users to freely navigate through different sections of the application. This makes it more intuitive for the users, especially as they gain the ability to redo actions such as choosing workouts. The application also has a search bar to search for specific exercises even though the system set the recommended exercise for the users. The app also requests users to start the exercise or set when the user is ready.





Consistency and Standards

The design layout is consistent across different sections of the app. This is accomplished by using

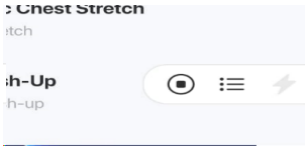


similar colour schemes and icons in order to not confuse the user. This consistent standard creates a more cohesive experience for the user.

Error Prevention

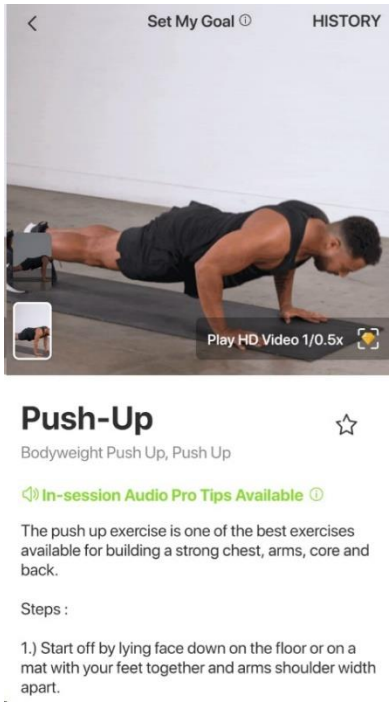
There are error messages used to prevent users from making mistakes. One such place this is integrated is during the workout session, where a pop-up message is used when users accidently exit the session. Furthermore, when exiting the ongoing session, the session is minimized while the timer is still on. This prevention tactic allows users to correct their mistake when made so. This comforts the user and builds confidence when browsing through the system.





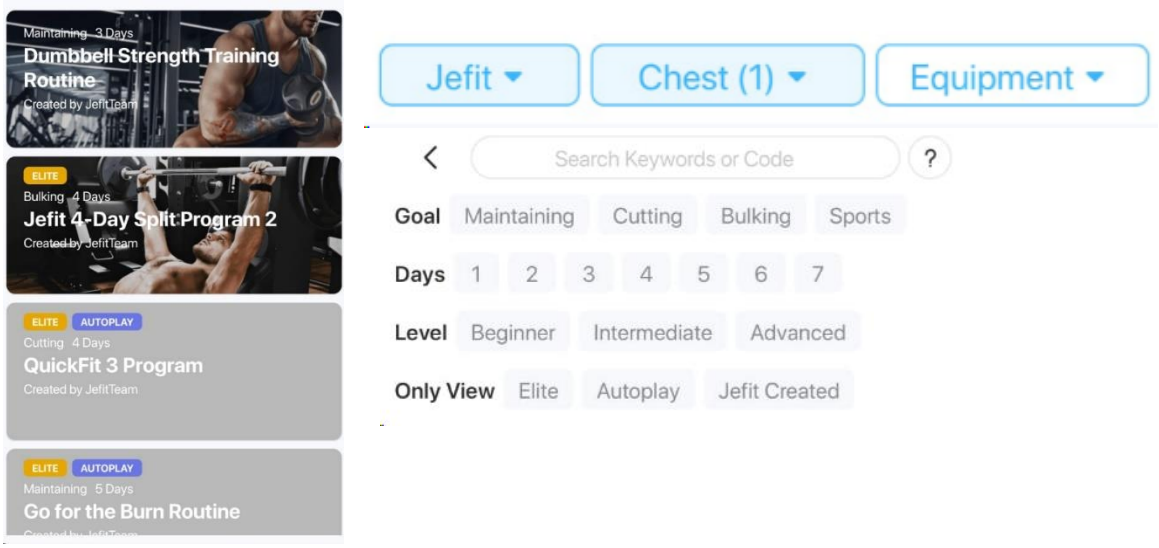
Recognition Rather than Recall

The app generally uses common icons, terminology and images that are easily recognizable, additionally the app also makes use of videos and photos to display the variety of exercises that the users can choose.



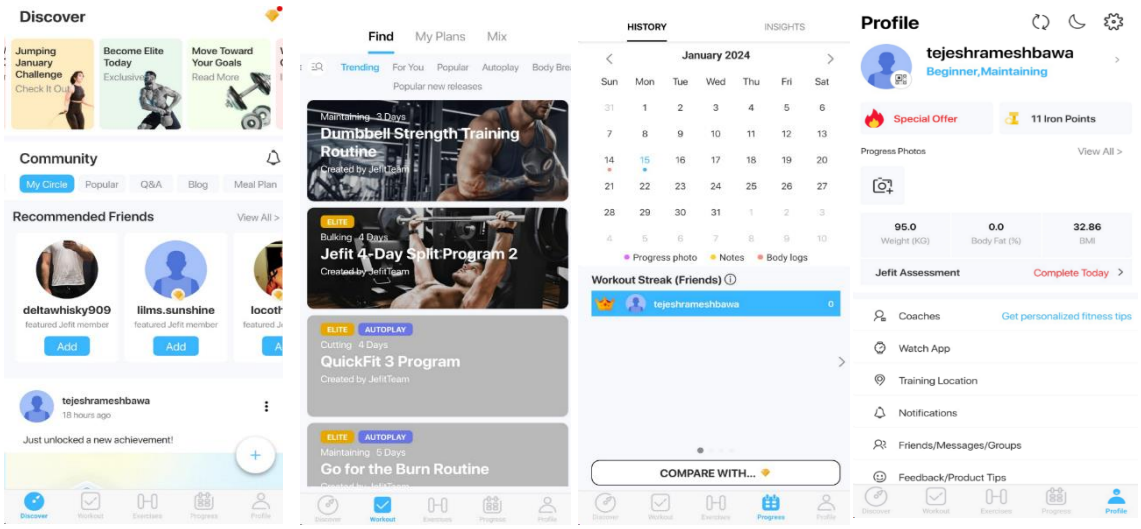
Flexibility and Efficiency of Use

The users are provided with customizable workouts even though the system provides tailored workout routines to users. However, this app gets very confusing for new users due to its complexity, but as the user get used to the app, it is rather easier to use the app.



Aesthetic and Minimalist Design

Jefit user interface design is generally clean and does not divert the users focus away from essential information. The app optimizes visual clarity however some pages are crowded which make it harder to find the required information. However, the efficiency of the app decreases because the system has many ads pop ups at the bottom of the page.

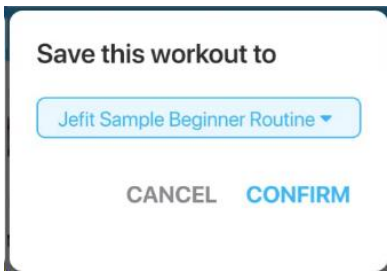


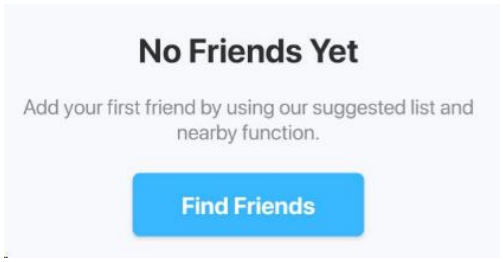


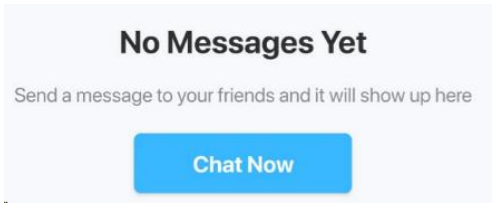


Help User Recognize, Diagnose, and Recover from Errors

The error messages generated are short and descript the problem in short. This is helping the user easily understand the problem with getting lost in the over explained message.

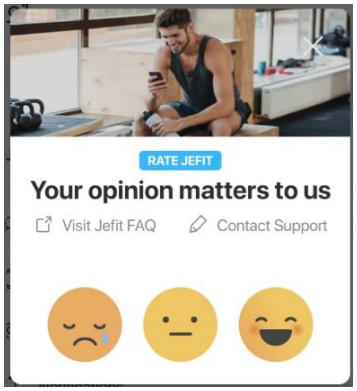






Help and Documentation

The application has a feedback/product tips button on the profile page which pops a message by providing a link to contact support and FAQs making it difficult to spot especially for new user. However, the online community provides an excellent accommodation to answer the user questions.



Heuristic	Fitbit	Zombies, Run!	Jefit
Visibility of System Status	9	8	9
Match Between System and the Real World	10	8	9
User Control and Freedom	9	7	8
Consistency and Standards	9	9	8
Error Prevention	9	8	8
Recognition Rather than Recall	8	7	9
Flexibility and	5	8	7

Efficiency of Use			
Aesthetic and Minimalistic Design	9	9	7
Help Users Recognise, Diagnose, and Recover from Errors	8	8	8
Help and Documentation	9	9	8
Total	**85**	**81**	**81**

3.5 Conclusion

In this evaluation, it can be noticed that Fitbit and Jefit perfectly fit the criteria of effective use of gamified elements. Although the most gamified elements incorporate in the app is Zombies, Run!, the system heavily focuses on the games storyline which therefore diverts from the main objective to motivate users to have a healthier lifestyle.

To conclude, this section provides with valuable insights to how applications in the market present different features and how they incorporate gamification elements into their respective product. Each application evaluated, display different styles, while making them similar to each other, they all have their distinct feature that make it unique in their respective field.

4. Requirement Analysis

4.1 Introduction

This section describes all high-level requirements to solidify expectations about what the system must, should, could, and will not. To support the development of the system, MoSCoW rules are used to prioritize high-level requirements before developing the system to ensure that the functional and non-functional requirements are clearly specified.

After carrying out extensive literature review and product research, it was confirmed that there were many users who lack motivation and were in dire need of some form of excitement to boost them to start or continue their workout and fitness journey. It was identified that many users wanted to gamify their workout routine, however the users had many functional requirements to consider the application satisfactory, this includes badges, points, ranking and social community.

The reason behind developing this app was personal experience, where it was noticed that many gamified fitness apps mainly focused on gamifying running, jogging and walking activities, and did not approach stationary exercise like push ups, sit ups, etc.

4.2 High-Level Requirements

- The system must provide all users with the functionality to login and create a profile.
- The system must provide all users to manage their profile and personal details.
- The system must allow users to choose various workout quests, view detailed description and complete quests.
- The system must reward points to users for completing quests by earning badges.
- The system must allow users to use social features and connect with other users.
- The system must allow users to operate a timer and what workout they are performing during their quest sessions.

- The system must auto update the points the user has earned in the app and database.
- The system must allow users to view all the badges they have earned as well as view all the badges they can possibly earn.
- The system must allow users to view their rank in the leaderboard as well as update according to the points they will earn.
- The system must ensure an intuitive and inclusive experiences for users of all fitness levels and abilities.
- The system must allow users to customize their workout routine.
- The system must ensure robust security measures protect user data, with clear communication of privacy policies and user content for data usage.
- The system must allow the admin to upload pdf documents on the web page which will be reflected on the application.
- The system must provide educational content on nutrition, fitness science and mental well-being to support user health through fitness professionals (admin).
- The system must allow the admin to regularly upload quests.
- The system must provide a 'contact me' page for users to send questions about health and fitness to the admin.
- The system must allow the user to download their fitness progress.
- The system must offer personalized workout recommendations based on user's preferences, fitness level and progress.
- The system must allow integration with fitness devices to sync data for enhanced tracking and insights.

4.3 MoSCoW Prioritising High-Level Requirements

No.	High-Level Requirement	Priority
1	The system must provide all users with the functionality to login and create a profile.	Must Have
2	The system must provide all users to manage their profile and personal details.	Must Have
3	The system must allow users to choose various workout quests, view detailed description and complete quests.	Must Have
4	The system must reward points to users for completing quests by earning badges.	Must Have
5	The system must allow users to operate a timer and what workout they are performing during their quest sessions.	Must Have
6	The system must allow users to view all the badges they have earned as well as view all the badges they can possibly earn.	Must Have
7	The system must allow users to view their rank in the leaderboard as well as update according to the points they will earn.	Must Have
8	The system must allow the admin to upload pdf documents on the web page which will be reflected on the application.	Must Have
9	The system must provide educational content on nutrition, fitness science and mental well-being to support user health through fitness professionals (admin).	Must Have
10	The system must allow the admin to regularly upload quests.	Must Have
11	The system must allow users to use social features and connect with other users.	Should Have
12	The system must auto update the points the user has earned in the app	Should Have

	and database.	
13	The system must ensure an intuitive and inclusive experiences for users of all fitness levels and abilities.	Should Have
14	The system must allow users to customize their workout routine.	Could Have
15	The system must ensure robust security measures protect user data, with clear communication of privacy policies and user content for data usage.	Could Have
16	The system must provide a 'contact me' page for users to send questions about health and fitness to the admin.	Could Have
17	The system must allow the user to download their fitness progress.	Won't Have
18	The system must offer personalized workout recommendations based on user's preferences, fitness level and progress.	Won't Have
19	The system must allow integration with fitness devices to sync data for enhanced tracking and insights.	Won't Have

4. 4 Determining and MoSCoW Prioritizing the Functional Requirements

The functional requirements will help determine how the actors will be able to communicate with the system. The actors will be users (individuals who will have access to the fitness app) and admins (individuals who are fitness professionals using the web page). By prioritizing the functional requirements would help to determine which requirements are appropriate for the development of the MVP while also considering for time constraints.

Actor	Requirements	Priority
User	User must be able to create or login to their profile and submit personal information.	Must Have
	User must be able to view and edit their personal information.	
	User must be able to logout to ensure that their profile is not accessible to other individuals.	
	Users must be able to view and select different workout quests to their needs which are set by the admins.	
	Users must be able to start, pause and end the timer during their workout sessions while being able to view the workout routine.	
	Users must be able to view their rank on the leaderboard.	
	Users must be able to view different admins (Fitness Professional) and can follow their social through the description they provide.	
	Users must be able to read educational content that the admin have posted.	
Admin	Admin must be able to create or login to their profile and submit personal information.	Must Have
	Admin must be able to view and edit their personal information.	
	Admin must be able to logout to ensure that their profile is not accessible to other individuals.	
	Admin must be able to add new workout quests.	
	Admin must be able to add new educational content.	
User	Users should have the ability to view and create public post to the community.	Should Have
	Users should be able to view the points they have earned.	
	Users should be able to view which badges they have earned with their points, and which badges they can earn in the future.	
User	Users could be able to customise their workout routine.	Could Have

	Users could be able to use the contact me button to contact the administrator about an issue they are facing with the application.	
User	Users could be able to download their data which is stored on the application.	Won't Have
	Users could be able to select a customized quest which is created by the system after learning the habits of the users.	
	Users could be able to sync their own fitness devices to the system.	
Admin	Admin could be able to delete inappropriate community posts.	Won't Have
	Admin could be able to delete and edit certain quests.	

4.5 Determining the Non-Functional Requirements

Non-Functional Requirements will help set a margin for the system to have sufficient quality. By addressing to these requirements, it makes the system functional and engaging and also reliable, secure and scalable.

4.5.1 Reliability

- The system will need to be optimized to give optimal performance by ensuring that users will have a seamless experience. This will include reducing load times for accessing quests. Community posts and educational resources. Additionally, implementing efficient database queries will help maintain availability under varying loads resulting in minimal downtime.
- The system should prioritize a smooth and intuitive user experience. This includes a responsive design for various devices, a clear navigation pathway and low latency during user interaction.

4.5.2 Security

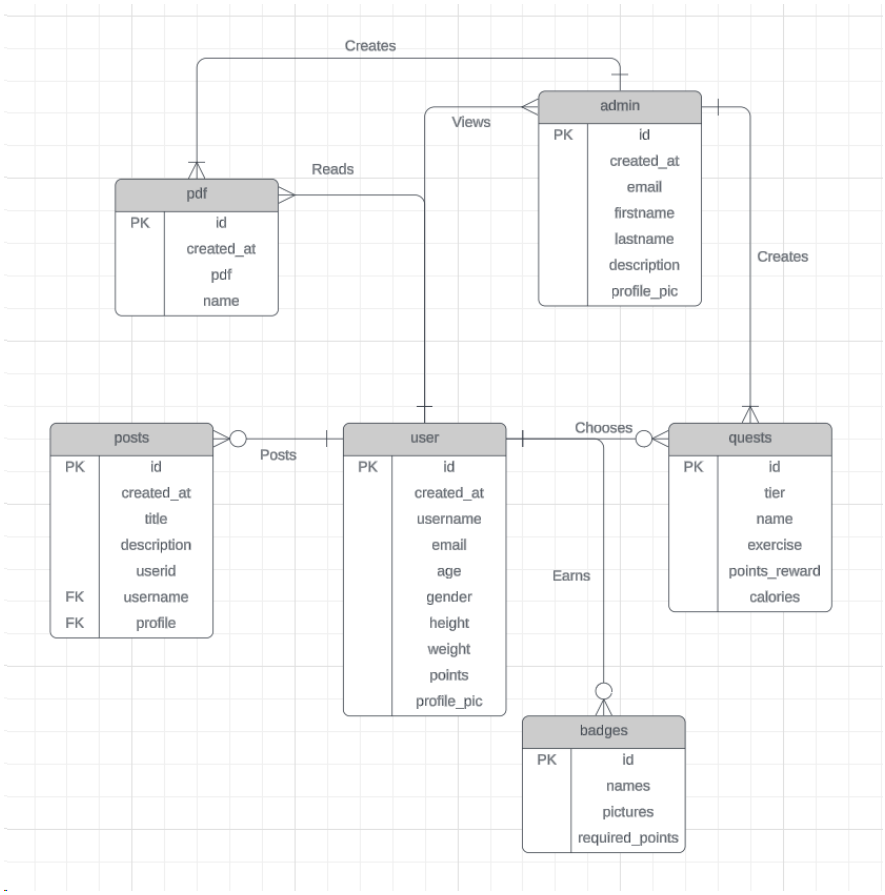
- It is imperative to enforce security measures in order to safeguard user data from unauthorised access or breaches. This can be accomplished by limiting the data access depending on user responsibilities, secure transmission methods and encrypting sensitive information.
- The system required to adhere to the DPA (Data Protection Act) and UK GDPR. This includes obtaining explicit user consent for processing user data and providing privacy policies and data storage policies. Furthermore, the system will avoid breaking any copyright laws which could create legal constraints.

4.5.3 Scalability

- The overall structure of the system should be adaptable and extensively documented to guarantee future scalability and easy maintenance. This allows functionality improvements and future upgrades to be made easier without interfering with the system in general.

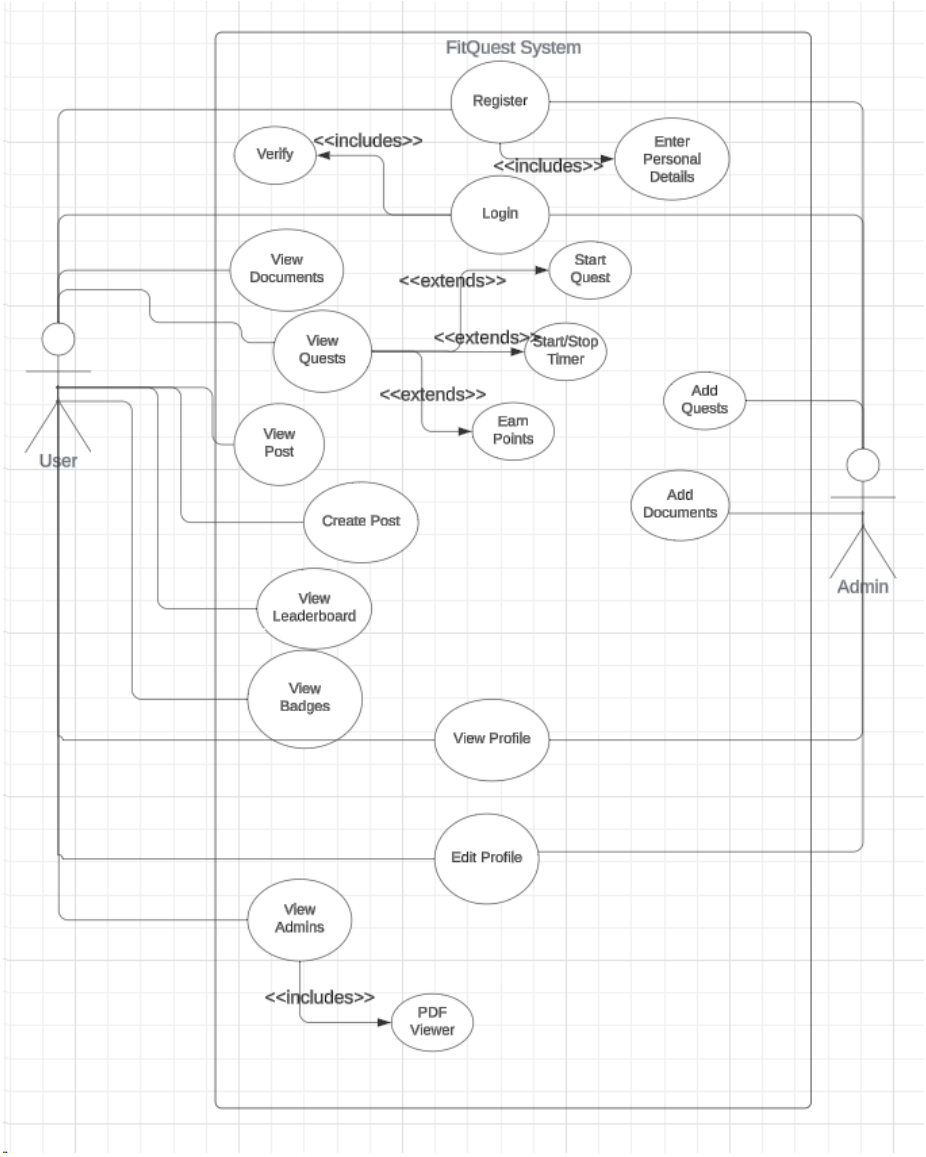
4.6 Entity Relationship Diagram

This section explores the details of the ERD that help understand and illustrate the relationship between entities in the project dataset. The ERD serves as a visual roadmap and helps clarify the connections and dependencies between entities. Thus, examining the ERD will help understand the data architecture in this project


Creates
admin
PK
id
created_at
email
firstname
lastname
description
profile_pic
Views
Reads
pdf
PK
id
created_at
pdf
name
Creates
posts
PK
id
created_at
title
description
userid
FK
username
FK
profile
Posts
user
PK
id
created_at
username
email
age
gender
height
weight
points
profile_pic
Chooses
quests
PK
id
tier
name
exercise
points_reward
calories
Earns
badges
PK
id
names
pictures
required_points

4.7 Use Case Diagram

The use case diagram below illustrates the interactions between the users, admin and the system. This Use Case Diagram helps to understand the layout a foundation, on how the system will operate considering the actions of the actor.



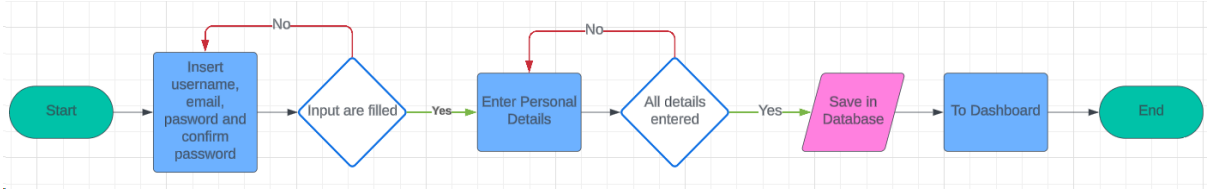
4.8 Activity Diagrams

Activity Diagrams in this section will help to understand how some of the functionalities will work in the system.

4.8.1 Registration of User

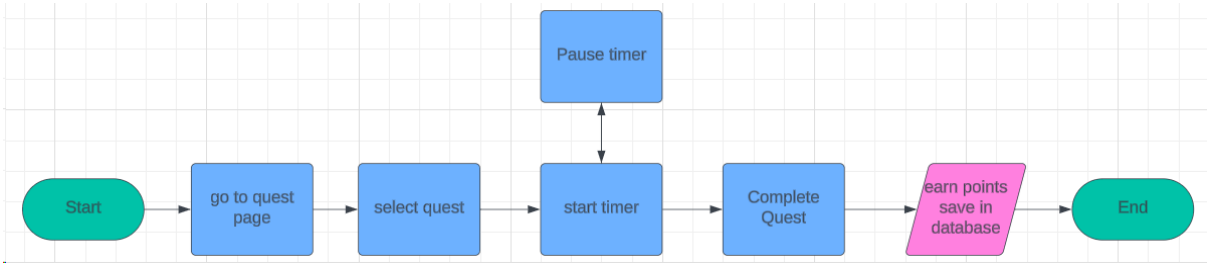
The image describes the registration of users. First, the user goes to the registration page, where he enters his username, email address and password and confirms this password. The input fields must be filled in order to move to the next page where the personal information is entered, which again requires the user to enter details. After that, their details will be saved in the data and the user will

be navigated to the dashboard.



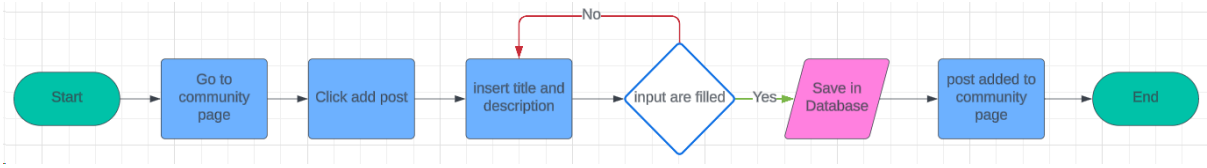
4.8.2 Users Starting a Quest

The users will start by going to the quest page, where they will select a quest of their liking. They will then start a timer which the user can pause and resume. After completing their workout, they will then complete the quest. And earn points which is saved on the database.



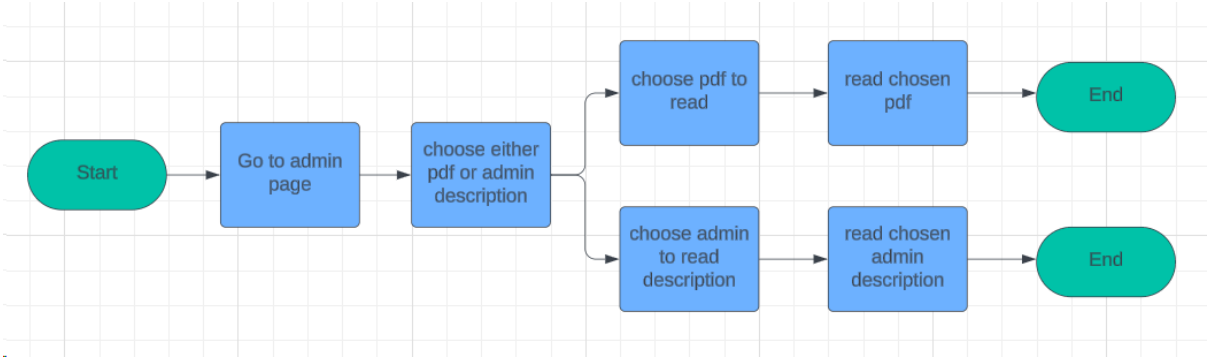
4.8.3 User Adding Posts on the Community page

The user will start by navigating to the community page, where they will click the add post button. They will then insert the input fills, which are required to be fill. The post is then saved in the database as well as posted on the community page.



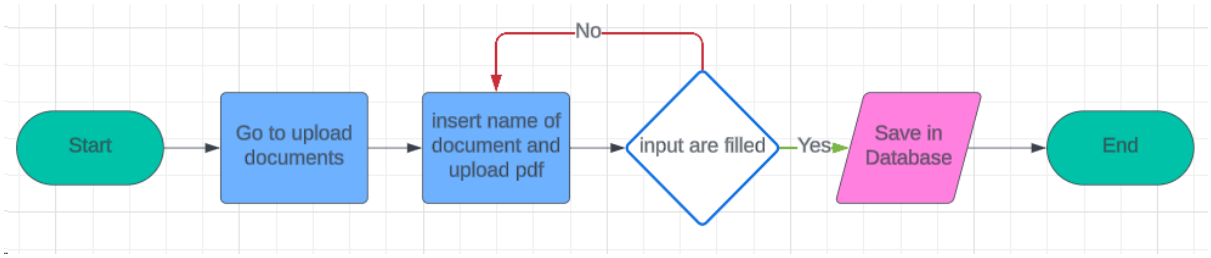
4.8.4 User Reading the Admin Description or PDF

The user will navigate to the admin page and will either choose to read a pdf or read the description by the admin.



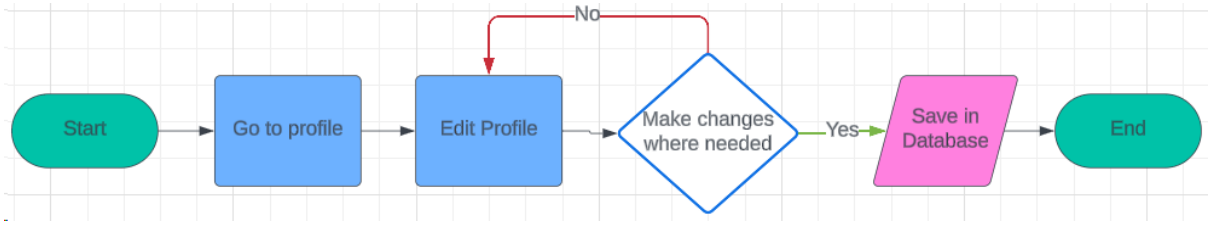
4.8.5 Admin Uploading PDF

The admin will navigate to the upload pdf page where they will have to write the name for the document and submit the document. If successful, the document will be saved into the database.



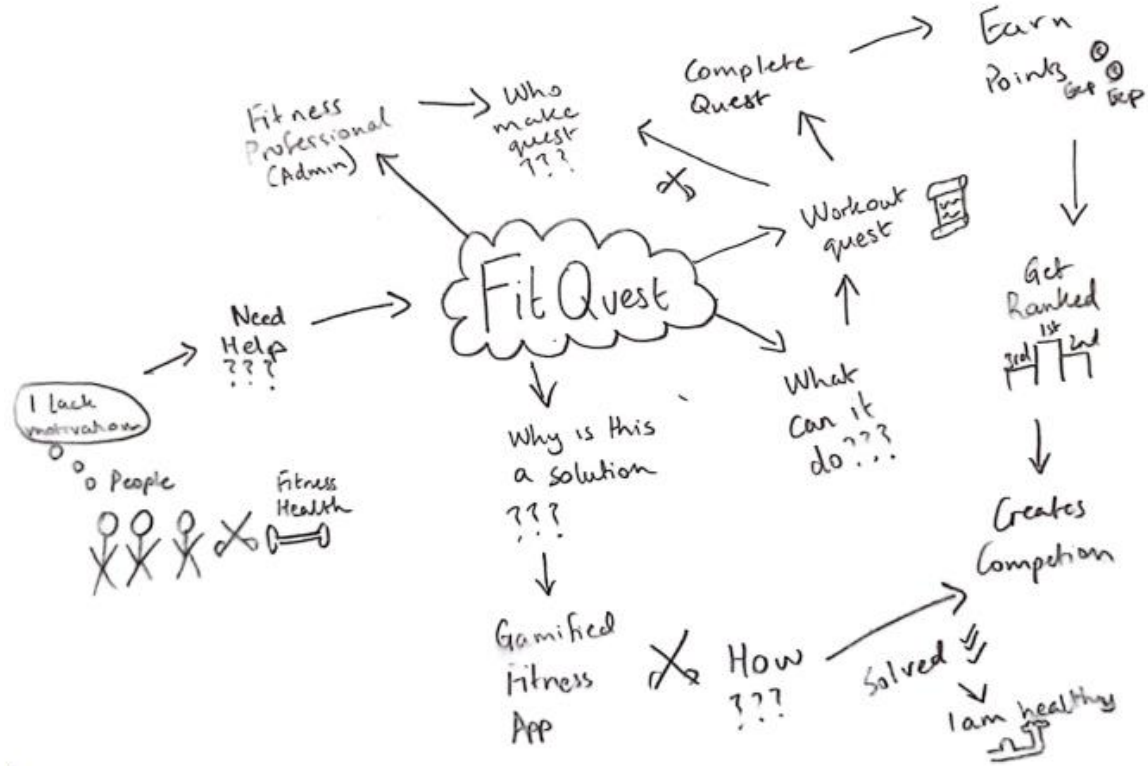
4.8.6 Admin Editing Profile

The admin will navigate to their profile page and press the edit profile button. Here they can make changes to their profile and updating it. This will then be saved in the database.



4.9 Rich Picture

The rich picture illustrates the main concept of the system FitQuest. This system will try to tackle the major motivational barrier that people face when they start their fitness journey or have already start their fitness journey. This system will make use of gamification to encourage users to continue their fitness journey. As illustrated the app will make the workout routines into quests which the user will have to complete to earn points. These points will also help them rank up in the leaderboard. The workout quest routines will be made by the admins who will provide their own descriptions with links to their social media which the user can read and follow them on there. The admin will also add educational content which will help the user understand more about their overall health and fitness.



5. Design Development

5.1 Introduction

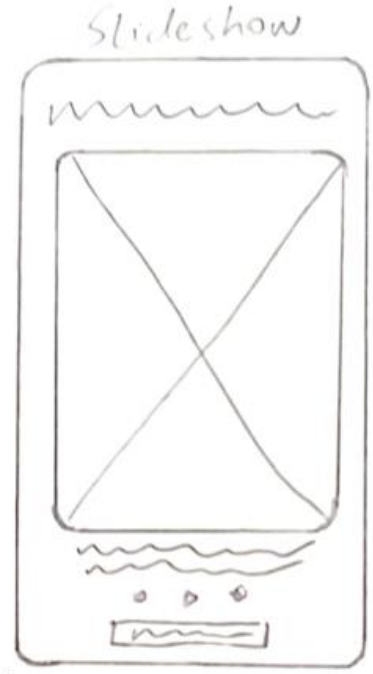
The initial release of the product will have a front-end design and back-end design. The user interface of the product will be as simple as possible making it accessible to as many individuals as possible. This section will explain how the wireframes were developed and how they will be implemented as a mid-fidelity prototype using Axure RP 10. To help with decision making during the design process, the rationale for the colour scheme and stylistic selections are also provided and supported by diagrams.

Furthermore, the design heuristics developed by Nielsen are crucial in the process of design development since they provide a clear yet broad set of principles for creating interfaces that are easy for users to navigate. By following these principles, the product will provide the requirements of users, reduce mistakes and promote a great user experience.

5.2 User Interface Wireframes

The finding from the literature review and requirement analysis, it provided the most optimal design for the system. These wireframes are made to help visualize how the UX design of the application will look like. The wireframes in this section will only include the main pages (More detailed explained are in the Appendix 2 Comparison of Wireframes, Mid-Fidelity and Final Product)

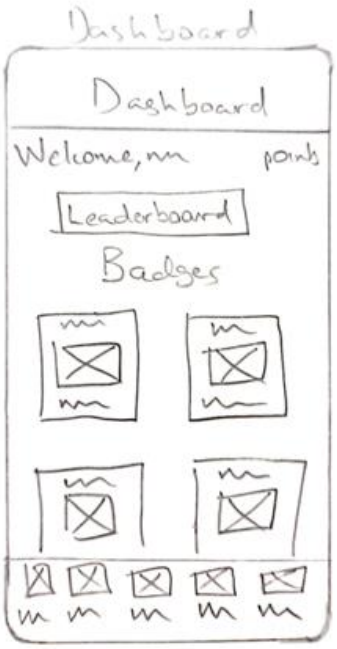
5.2.1 User Application



The users are welcomed to the app, this page includes a slideshow. This will allow the user to expect what this app is about.



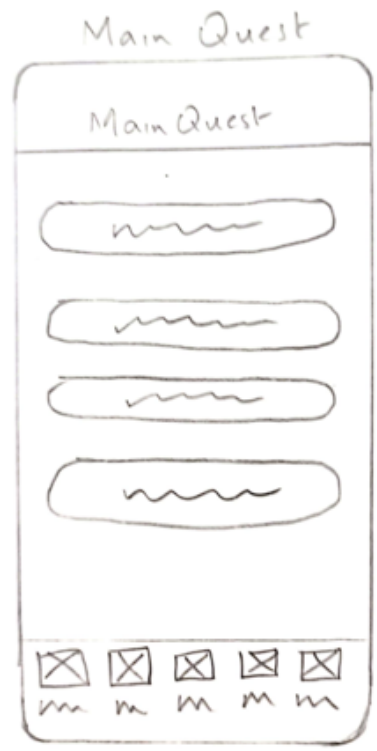
The users will then navigate to the Login Page where they will have to Login or Register a new account.



The users on the dashboard can view their points and badges and navigate to the leaderboard.



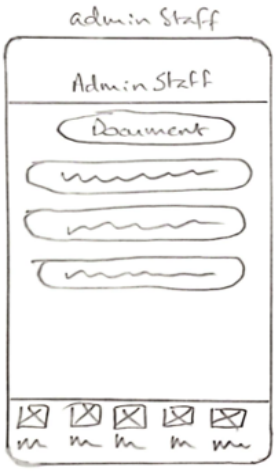
The user in this page can view their profile or status. On this page the user will be able to go to the settings and also update their profile picture.



On this page the user will be able to choose the difficulty of their quest and consider which quest they want to complete depending on the difficulty.

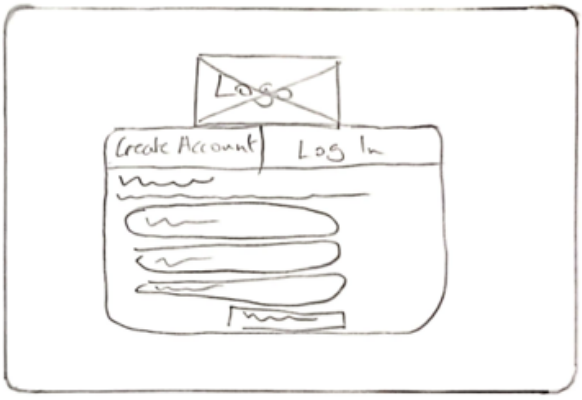


On the community page the user will be able to view all the posts posted by other users as well as navigate to the add post page.

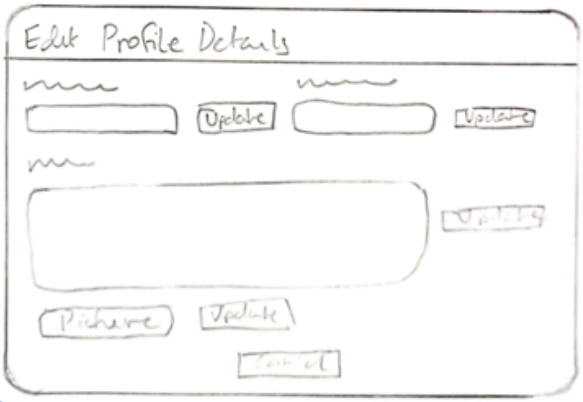


This page shows the admin staff, where the user can either select to view their profile or navigate to read documents and articles.

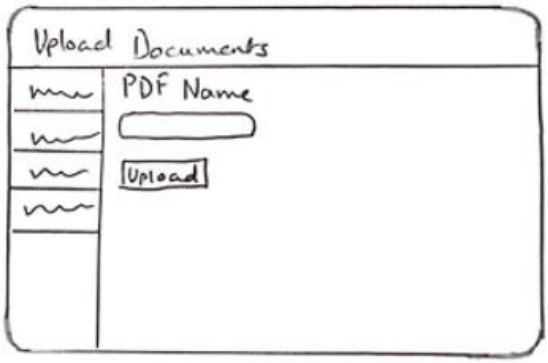
5.2.2 Admin Application



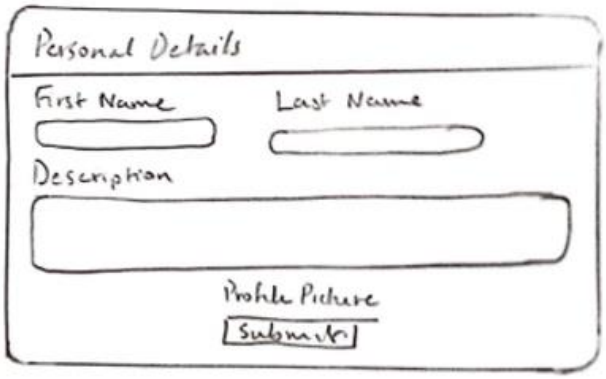
This page a web page for the admin where they can log in or register their account.



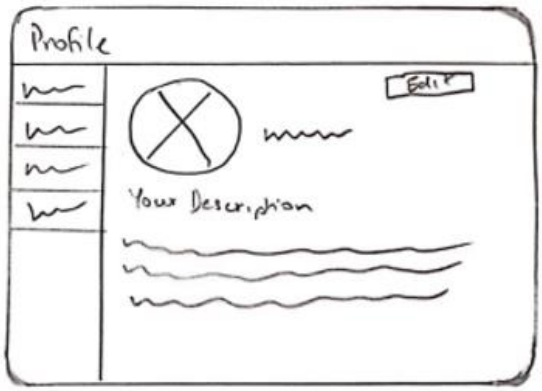
On this page the users will be able to edit their profile details



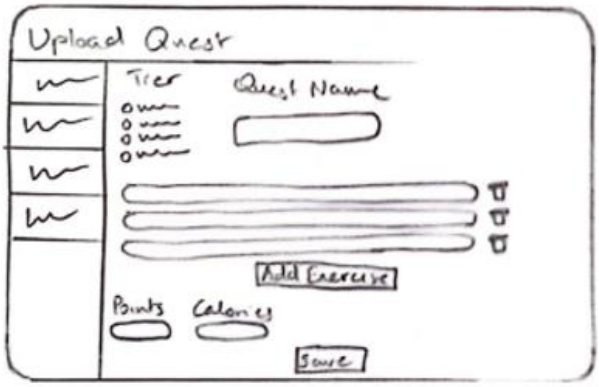
This page will allow the admin to upload any piece of content, which is in PDF format.



This the page the admin will navigate to when they have created a new account. They will have to input their personal details.



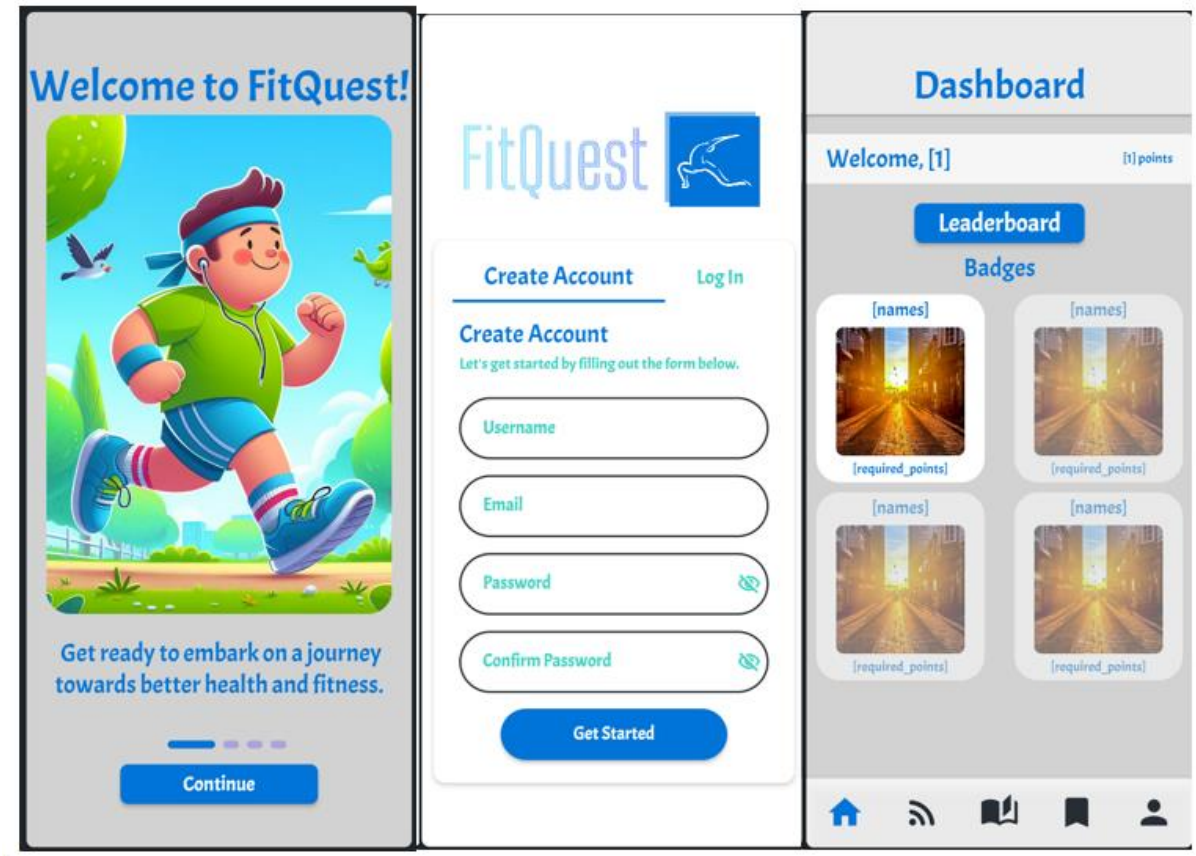
On this page the admin can view their profile. This page also will have an edit button to navigate the admin to edit their profile.

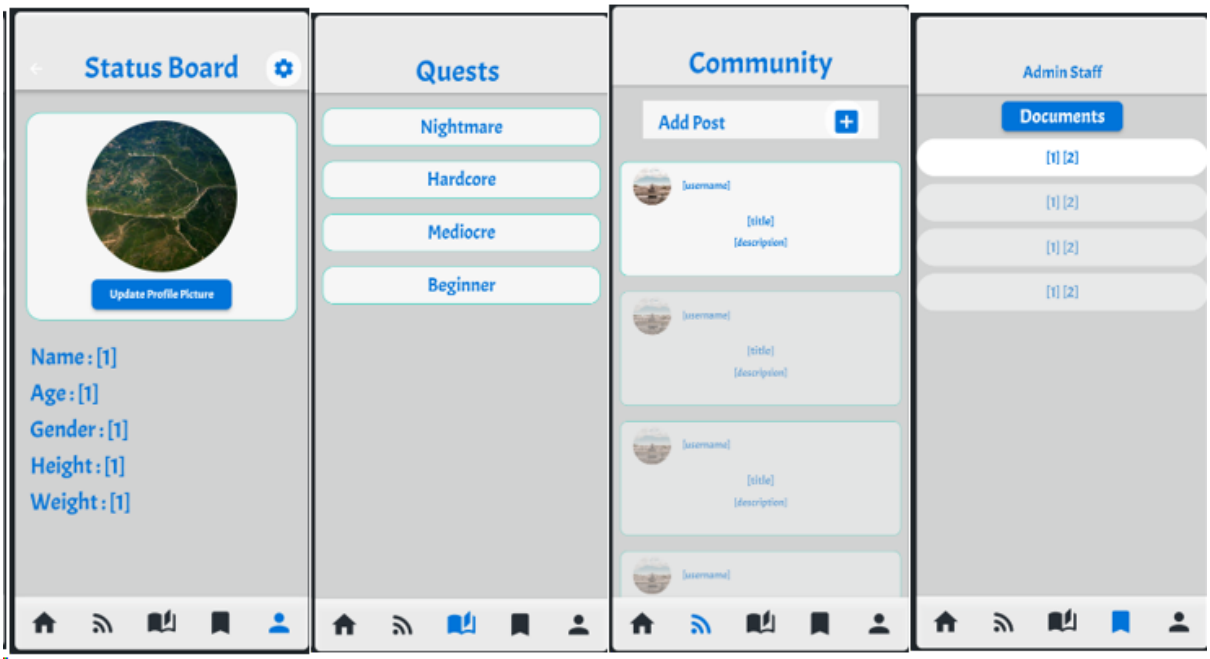


On this page the admin will be able to upload new quest for the users

5.3 Mid-Fidelity Prototype

After designing the wireframes, the designs were then implemented into mid-fidelity prototypes using the Flutterflow Application





5.4 Design Elements

5.4.1 Colour Scheme



The purpose of this design is the simple and minimalist colour scheme is to maximize user engagement and experience. The strong blue colour of the Primary Text (#0074d9) contrasted with the light Primary Background (#ccffffff) makes important information easy to see and read, making it easier for users to navigate their fitness journey. The design maintains consistency and clarity, helped by the mild teal colour of the Secondary Text (#39d2c0), which provides a little contrast without overpowering the main information. Meanwhile, the white Secondary Background (#ffffff) draws attention to the interactive parts and expands the sense of space, improving the overall visual appeal. These well-chosen colours not only provide an aesthetically pleasing interface, but also make it easier for users to interact, increasing the effectiveness and fun of the training experience.

5.4.2 Fonts

Acme's strong dynamic properties make it a suitable font for this system. Its rounded edges and clean, clear lines give the impression of movement, which corresponds exactly to the motivating nature of fitness exercises. Additionally, Acme's readability ensures that users can quickly understand instructions and prompts, improving their overall experience. Acme helps create a visually appealing and intuitive user interface that encourages continued engagement and development within the system by combining style and clarity.

6. Technical Project Development

6.1 Introduction

The system was developed using FlutterFlow and Supabase. This creative platform combines a robust backend functionality with simplified programming. By leveraging FlutterFlow's adaptability, the system is able to create a smooth user experience across multiple devices. FlutterFlow has a powerful component library with an easy-to-use visual interface that accelerates prototyping and iterations. In addition to the powerful front-end tool, the backend is managed by Supabase, a dynamic open-source platform that provides a secure authentication service, scalable databases and real-time processing. The system guarantees data integrity, speed optimization and smooth data synchronization functions. By working together, Flutterflow and Supabase have enabled the creation of a comprehensive fitness system that puts user experience first.

6.2 Flutterflow

With absolutely no experience in coding apps, FlutterFlow was a great help in visualizing and implementing the functionalities of the system. FlutterFlow is a visual interface builder that is built on top of Flutter, Google’s open-source UI toolkit. FlutterFlow allows to drag and drop widgets and eliminates the need to manual code. It seamlessly translates these visual designs into clean and efficient Dart codes.

6.3 Supabase

Supabase is an open-source alternative to the traditional backend-as-a-service (BaaS) providers and offers a multitude of tools and services such as database management, authentication and real-time handling. Supabase also offers PostgreSQL, which make easy to understand SQL interfaces. Supabase powerful authentication mechanisms which is used in this project allow to automatically authenticate the users email and password. It can be stated that Supabase is a scalable, reliable and secure modern backend platform.

6.4 Changes Made

The major functionalities in the system were implemented well. All the major pages were designed appropriately. However, when it came to the navigation bar in the user app, the design includes text of page's name under the icon. This was not possible because it would cause a cluster and would divert the user's attention away from the main focus of the page. Therefore, removing that text created a better look for the system.

In the initial development (before the design phase), there wasn't any needs for an admin page. However, considering the importance of having professional fitness influencers on the app could help the users to trust the app. The admin webpage was developed after taking that into consideration.

6.5 Technical Problems Faced

The development of the gamified fitness app had many technical problems. However, due do a meticulous project scheduling which helped mitigate any time constraints. When starting to develop the system, the major setback which was faced was when the system was not save in FlutterFlow and lost all the build system. However, the good thing was that the system that had been developed was just started and had barely 2 or 3 pages built.

One of the main problems faced was that when the user adds a null value in any of the input full, it is passed without verifying that input fill needs to be filled. This can be noticed when the user inputs in the personal detail page, the post page and in the update personal details page, also the same thing happens in the admin webpage in the personal details page, document uploads and add quest page. This is still not fixed yet. The admin webpage did not have any other major problem apart from this.

The admin in this system will not be able to access the database and system code and will only be able to use the webpage and if an issue arises, they will have to contact the system administrators for any issues.

7. Legal, Social, Ethical and Professional Issues

7.1 Legal

During the development of the system, it is crucial to carefully evaluate and deal with different legal aspects in order to guarantee adherence to relevant laws and regulations, namely the Data Protection Act 2018 and the General Data Protection Regulation (GDPR). These rules establish strict criteria for safeguarding user data and privacy rights, and non-compliance can lead to harsh consequences.

In accordance with the Data Protection Act 2018, the system is required to have valid legal reasons to collect and analyse user data, which usually requires obtaining permission from users. This permission must be voluntary, clear and well-informed and if the user wants to revoke their consent,

they should be able to do it at any point of time. Furthermore, the system has to give users transparent information on how their data is gathered, saved and handled, which has to be done by providing the users with terms of services or privacy policy.

Adhering to the General Data Protection Regulation (GDPR) is essential as it offers a thorough guideline for safeguarding user rights. This involves the ability to view personal data, correct errors, remove data (sometimes referred to as the “right to be forgotten”), limit or object to specific processing. The system has to implement various features to support the rights of the users such as a way to modify and gain access to their personal data or requesting to deleting their data. The Equality Act of 2010 which prohibits discrimination based on a person’s race, sex, or disabilities, is yet another law that applies to this project. The application should not upset the users because of their ethnicity, sex, or disabilities and instead make the users feel comfortable to use the app.

It becomes clear that upholding intellectual property rights is essential for complying with the law. Obtaining the necessary licenses or permits to use third party content, like PDFs, pictures or exercise programs. This will help prevent a copyright infringement lawsuit. By respecting the UK copyright laws, it will legally guarantee that the material used to develop this app will have the owner’s intellectual rights. Moreover, it is essential to create a clear and open binding agreement with users, staff, and any outside service providers involved in the administration of the application. by outlining the rights, responsibilities and procedures for resolving any disputes, these agreements will help promote a sense of responsibility among all parties that are involved and reduce legal uncertainty.

Overall, the system may function within legal bounds while protecting user data and rights by abiding by these regulatory structures and placing forth the necessary security measures. This promotes confidence and trust among users while also guaranteeing compliance with regulatory standards, creating a safe and supportive atmosphere for the user’s fitness journey.

7.2 Social

It will be necessary to consider a wide range of social issues when creating this system to ensure that it is user-friendly and promotes a sense of community and inclusion. Cultural sensitivity plays a crucial role in acknowledging the diverse cultural backgrounds and viewpoints of its users. To create a pleasurable environment for all users, requires the development of engaging strategies that honours and complement various cultural norms and values.

Additionally, the system will proactively encourage community participation in line with public health initiatives. The system will provide a sense of community and social support by encouraging user interactions and collaboration. This community-driven strategy not only improves the overall user experience, but also encourages people to exercise and work together on their fitness goals. The system creates an agile platform that constantly connects and succeed together by fostering a sense of solidarity and shared purpose.

To meet the needs of users with different backgrounds, skills and interests, the system also places great emphasis on accessibility and inclusion. This includes offering support for multiple languages, implementing features that make materials accessible to people with disabilities, and ensuring that activities and content are appropriate and take cultural differences into account. The system’s diversity and inclusion increase user satisfaction and engagement by creating a vibrant and encouraging community that gives all users the confidence and respect to achieve their fitness goals.

In conclusion, by considering social factors such as inclusion, community involvement and cultural sensitivity, the system can develop a platform that promotes not only physical exercise but also a

sense of empowerment, unity and belonging among its diverse user bases . The approach improves the user experience, promotes social well-being and supports the overall success of the platform.

7.3 Ethical

The development of this system requires careful consideration of ethical principles, particularly with regard to user privacy and the impact of gamification features. Transparency and respect for users' data rights form the central ethical framework. To achieve this, transparency in the collection, storage and processing of data must be a top priority. Respecting user rights is not only required by law and morality, but also ethically.

When developing the system, it is important to be careful not to inadvertently encourage dangerous or extreme training programs when incorporating gamification elements. Although there is a possibility that gamification could encourage people to take actions that could be detrimental to their well-being, it also has the potential to be a very effective tool for encouragement. In order to promote lasting and healthy exercise habits, the ideal balance must be found rather than pushing people to their limits or encouraging unhealthy competition.

The system also needs to consider the moral impact of user participation and their mental health. When developing the system, it is crucial to consider the potential impact of game elements such as leaderboards, badges and points on players motivation, self-esteem and overall mental health. Prioritizing user's long-term well-being and encouraging positive behaviours that support their overall fitness and health goals should be a top priority.

By prioritizing these ethical factors, the system can create a comprehensive and encouraging environment that helps users achieve their fitness goals. This includes compliance with ethical guidelines, which, in addition to compliance with the law, also put the privacy, data protection and well-being of users first. By encouraging healthy behaviours, transparency, and recognition of user rights, the system can create an environment that is both positive and encouraging for users to interact with and benefit from.

7.4 Professional

Developers of a system must adhere to professional standards and practices as set out in the BCS Code of Conduct. The foundation of professionalism will be cybersecurity and quality assurance throughout the development process. Extensive security audits, regular penetration testing, and implementation of strict quality assurance procedures are necessary to protect user data and build and maintain user trust.

Maintaining professionalism also includes adhering to compliance requirements and regulatory updates. It is essential for developers to stay informed about the ever changing legal and regulatory environment, especially when it comes to privacy and data security. It is not only a technical but also a legal obligation to adapt the system to the latest serious consequences, including fines, legal consequences and reputational damage. Therefore, to ensure the integrity and legality of the system, it is essential to maintain compliance with regulatory standards.

In addition, professionalism is maintained even after the system is launched through regular maintenance and support. Any security vulnerabilities or issues that may develop over time must be addressed by developers through timely updates and patches. Additionally, to ensure that the system keeps pace with the changing needs and expectations of its users, user feedback should be actively sought and incorporated into the development roadmap. By maintaining these professional

standards, the system can continue to conform to best industrial practices while providing users with a reliable and high-quality platform.

The system can provide users with a reliable and high-quality platform while adhering to best industrial practices if it considers these professional considerations and follows the guidelines set out in the BCS Code of Conduct. This shows a commitment to morality and responsibility in the development and application of technological solutions, which not only strengthens users' trust and confidence.

8. Product Testing

To ensure that the developed product is functional and ready for use, testing is a crucial step in the project completion process. To ensure a thorough review, the details of the assessment are listed in the table below. In structured testing, the tasks are selected based on the specifications from the "Requirements Analysis" section.

No.	What is being tested	How and Where	Expected Result	Actual Result	Action Taken
1	User must be able to create or login to their profile and submit personal information.	Users must be able to login by entering their email and password or create a new profile by entering their email, password and confirming their password. On the Auth2 page.	User is successfully able to either login or create a new profile	Successfully login and creation of new profile	None
2	User must be able to view and edit their personal information.	By editing their personal information which was taken when creating the account. On the Update Information Page, via the setting page	The personal information was successfully updated	The information was updated	None
3	User must be able to logout to ensure that their profile is not accessible to other individuals.	By pressing the logout button in the setting page	Successfully logging out of the app and displaying the Auth2 page	Logged out and on the Auth2 page	None
4	Users must be able to view and select different workout quests to their needs which are set by the admins.	By going to the mainquest page which shows quests which are categorized by difficulty. The user will have to choose the which difficulty they want and then will be able to start the quest.	User will successfully be able to choose the difficulty and quest within it.	The quest was not shown according to their difficult and displayed all the quest.	There was a problem in the page parameter, as the correct parameter was not passed in the mainquest page.
5	Users must be	After choosing the	User will	The timer	Simple fix was

	able to start, pause and end the timer during their workout sessions while being able to view the workout routine.	quest, the user can start and pause and end their timer.	successfully start the timer and when pause the end session button will disappear and when resumed it will appear again. The end session will complete the exercise and the user will earn the points.	setting was displayed properly. But when the session is ended the points earned would reset to the points earned to that quest and not add on.	made. A code expression was passed to add the current points the user had, and the points earned from the quest.
6	Users must be able to view their rank on the leaderboard.	By going to the leaderboard page, the user will be able to view their rank.	User will successfully see what rank they are.	The ranking system was implemented well. However, the index started on zero instead of 1	Simply added an expression code by having the index added to one.
7	Users must be able to view different admins (Fitness Professional) and can follow their social through the description they provide.	The user is able to view different types of admins and click on them to view their profile. This will happen on the admin page.	The user is successfully able to view the admin profile.	The admin page was working well, with the ability to select each admin and capabilities to view their profile.	None
8	Users must be able to read educational content that the admin have posted.	The user is able to select the documents button on the admin page to view all the documents posted by the admin. The user is also able to select each document and read it in a new page.	The user is successfully able to view all necessary documents.	The documents functionality worked properly and navigated to the documents page. The documents were selectable and accessible.	None
9	Admin must be able to create or login to	Admin must be able to login by entering their email and password or	User is successfully able to either	Successfully login and creation of	None

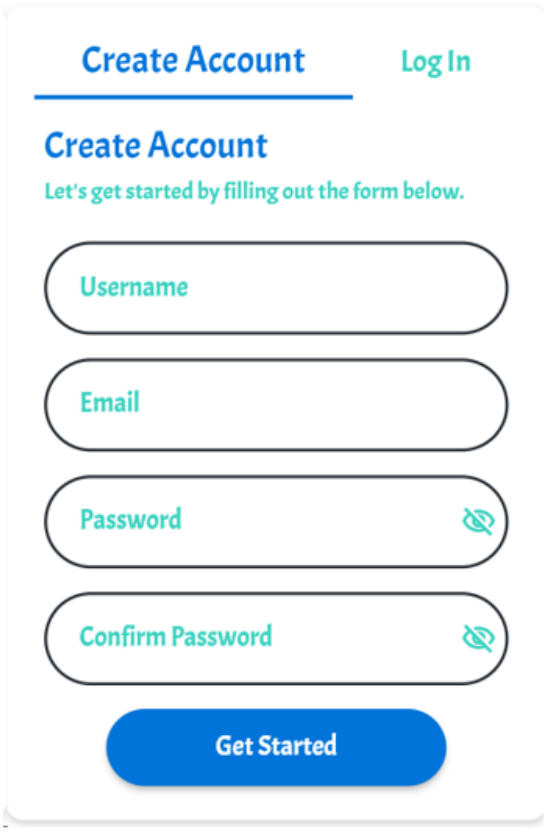
	their profile and submit personal information.	create a new profile by entering their email, password and confirming their password. On the Auth2 page.	login or create a new profile	new profile	
10	Admin must be able to view and edit their personal information.	By editing their personal information which was taken when creating the account. On the Profile Page, via the edit button	The personal information was successfully updated	The information was updated	None
11	Admin must be able to logout to ensure that their profile is not accessible to other individuals.	By pressing the logout button in the setting page	Successfully logging out of the app and displaying the Auth2 page	Logged out and on the Auth2 page	None
12	Admin must be able to add new workout quests.	On the upload quest page, the admin will fill all the input field and press the save button.	The quest is successfully saved in the database and reflected on the user app.	The quest was uploaded and saved.	None
13	Admin must be able to add new educational content.	On the upload documents page, the admin will fill all the input field and press the upload PDF button to upload the document.	The document is successfully saved in the database and reflected on the user app.	The document was uploaded and saved.	None
14	Users should have the ability to view and create public post to the community.	On the community page the user is able to view all the post made by other users. The user can then press the add post button navigate to the add post page to fill in the input field and post their post.	The user is successfully able to view all community post as well as make post to the community.	The community page displayed all the post and was able to make posts	None
15	Users should be able to view the points they have earned.	On the dashboard page the user is able to see the points that they have earned.	The points are successfully displayed and updated on the page.	The points are correctly displayed.	None
16	Users should be able to view which badges they have earned with their points, and which	On the dashboard page the user us able to view all the badges that they have earned as well as all the badges that they can possibly earn.	The badges are displayed by showing which badges they have earned by varying the	The badges earned were shown correctly.	None

	badges they can earn in the future.		opacity.		

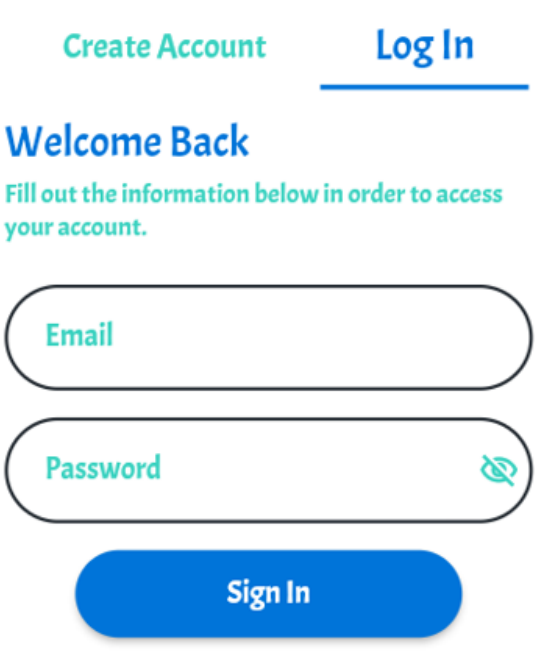
9. Implementation

9.1 User Application

```
Align(
  alignment: AlignmentDirectional(0, 0),
  child: Padding(
    padding: EdgeInsetsDirectional.fromSTEB(0, 0, 0, 16),
    child: FFButtonWidget(
      onPressed: () async {
        GoRouter.of(context).prepareAuthEvent();
        if (_model.passwordCreateTextController.text !=
            _model.passwordConfirmTextController.text) {
          ScaffoldMessenger.of(context).showSnackBar(
            SnackBar(
              content: Text(
                'Passwords don\'t match!',
              ),
            ),
          );
          return;
        }
        final user = await authManager.createAccountWithEmail(
          context,
          _model.emailAddressCreateTextController.text,
          _model.passwordCreateTextController.text,
        );
        if (user == null) {
          return;
        }
        await UserTable().insert({
          'id': currentUserUid,
          'created_at': supaSerialize<DateTime>(getCurrentTimestamp),
          'username': _model.userNameTextController.text,
          'email': _model.emailAddressCreateTextController.text,
        });
        context.pushNamedAuth('PersonalDetails', context.mounted);
      },
      text: 'Get Started',
```




```
Align(
  alignment: AlignmentDirectional(0, 0),
  child: Padding(
    padding: EdgeInsetsDirectional.fromSTEB(0, 0, 0, 16),
    child: FFButtonWidget(
      onPressed: () async {
        GoRouter.of(context).prepareAuthEvent();
        final user = await authManager.signInWithEmail(
          context,
          _model.emailAddressTextController.text,
          _model.passwordTextController.text,
        );
        if (user == null) {
          return;
        }
        context.goNamedAuth('HomePage', context.mounted);
      },
      text: 'Sign In',
      options: FFButtonOptions(
        width: 230,
        height: 52,
        padding: EdgeInsetsDirectional.fromSTEB(0, 0, 0, 0),
        iconPadding: EdgeInsetsDirectional.fromSTEB(0, 0, 0, 0),
        color: FlutterFlowTheme.of(context).primary,
        textStyle: FlutterFlowTheme.of(context).titleSmall.override(
              fontFamily: 'Acme',
              color: Colors.white,
              fontSize: 18,
              letterSpacing: 0,
            ),
        elevation: 3,
        borderSide: BorderSide(
          color: Colors.transparent,
          width: 1,
        ),
        borderRadius: BorderRadius.circular(40),
```



```
await UserTable().update(
  data: {
    'age': int.tryParse(_model.ageTextFieldTextController.text),
    'gender': _model.genderTextFieldTextController.text,
    'height':
        int.tryParse(_model.heightTextFieldTextController.text),
    'weight':
        int.tryParse(_model.weightTextFieldTextController.text),
    'profile_pic': _model.uploadedFileUrl,
  },
  matchingRows: (rows) => rows.eq(
    'id',
    currentUserUid,
  ),
);
context.goNamed('HomePage');
```



	cr... ti...	user... t...	email text	age int2	gen... t...	hei... i...	wei... i...	poi... i...	profile_pic text
ff3a1ea8-e95e-	2024-04-13 17:4	why	why@gmail.com	34	Male	324	213	135	https://kdqgpuxlaict
130b5711-2ae2-	2024-04-11 21:13	what	what@gmail.com	22	Female	22	22	2	https://kdqgpuxlaict
a95db465-46b	2024-04-11 23:2	trying	trying@gmail.com	50	Male	170	99	120	https://kdqgpuxlaict
056a1862-e60c	2024-04-10 21:0	testy	test@gmail.com	43	Male	175	75	743	https://kdqgpuxlaict
72f29b91-8b64	2024-04-14 17:0	okokok	ok@gmail.com	213	Male	123	123	562	https://kdqgpuxlaict

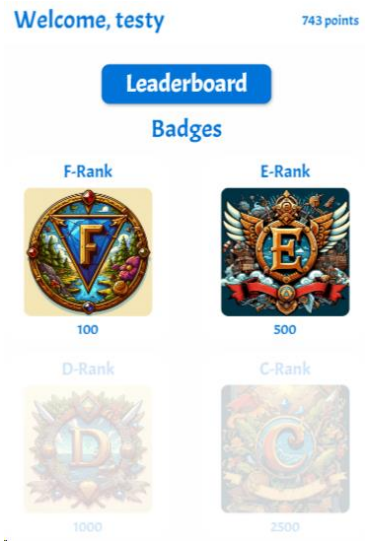
The code snippet above showcases the functionalities of the buttons on the Login page and the personal details page. When the user is registering for the first time, the user will have to enter the required field and will be then navigated to the personal details page to get relevant information. This can be confirmed on the database that all created users are successfully saved in the database. If the user requires to log in, they will be authenticated via the Supabase Authentication service and will then automatically navigate to the dashboard.

```
child: FutureBuilder<List<UserRow>>(
  future: UserTable().querySingleRow(
    queryFn: (q) => q.eq(
      'id',
      currentUserUid,
```

```
Text(
  'Welcome, ${rowUserRow?.username}',
  style: FlutterFlowTheme.of(context).bodyMedium.override(
        fontFamily: 'Acme',
        fontSize: 25,
        letterSpacing: 0,
      ),
```

```
Text(
  '${rowUserRow?.points.toString()} points',
  style: FlutterFlowTheme.of(context).bodyMedium.override(
        fontFamily: 'Acme',
        letterSpacing: 0,
      ),
```

```
Opacity(
  opacity:
      columnUserRow!.points! >= gridViewBadgesRow.requiredPoints! ? 1.0 : 0.2,
```



The code snippet above showcases the functionality call the users username and the points they currently have. The first snippet shows how the system will first compare the users authentication with the primary id from the user table to learn which user is logged in. It will then compare the ids and display the results on the screen. Furthermore, the opacity of the badges have been set in such a way that when the users does not have sufficient points the opacity of the badge would be 0.2 and and earned, it would be 1.

```
Builder(
  builder: (context) {
    final exercise = widget.quest?.exercise?.toList() ?? [];
    return Column(
      mainAxisSize: MainAxisSize.max,
      children: List.generate(exercise.length, (exerciseIndex) {
        final exerciseItem = exercise[exerciseIndex];
        return Stack(
          children: [
            Align(
              alignment: AlignmentDirectional(0, 0),
              child: Padding(
                padding: EdgeInsetsDirectional.fromSTEB(0, 20, 0, 0),
                child: Text(
                  exerciseItem,
                  style: FlutterFlowTheme.of(context).bodyMedium.override(
                        fontFamily: 'Acme',
                        fontSize: 25,
                        letterSpacing: 0,
```



nightmare	One Punch Man	["1","2","3","4"]

When the user chooses a quest it will display the exercises in that quest. The code shows that it had to pass a page parameter in the pervious page to learn the choice of the user and to display the correct execises. In this instance the nightmare difficulty was choosen and the One Punch Man quest was choosen. The exercises from the database are being successfully displayed on the screen.

```
List<UserRow> buttonUserRowList = snapshot.data!;
final buttonUserRow =
    buttonUserRowList.isNotEmpty ? buttonUserRowList.first : null;
return FFButtonWidget(
  onPressed: () async {
    setState(() {
      _model.add = _model.add +
          ((widget.quest!.pointsReward!) + (buttonUserRow!.points!));
    });
    await UserTable().update(
      data: {
        'points': _model.add,
      },
      matchingRows: (rows) => rows.eq(
        'id',
        currentUserUid,
      ),
    );
    ScaffoldMessenger.of(context).showSnackBar(
      SnackBar(
        content: Text(
          'Congratulations 🎉🎉🎉, you have earned${widget.quest?.pointsReward?.toString()}',
          style: TextStyle(
            color: FlutterFlowTheme.of(context).primaryText,
          ),
        ),
        duration: Duration(milliseconds: 2000),
        backgroundColor: FlutterFlowTheme.of(context).secondary,
      ),
    );
    context.goNamed('HomePage');
```



This snippet involves completing the quest. After completing the quest, the points earned by user are added to the existing points of the user. Additionally, a snack bar pops up congratulating the user.

```
Row(
  mainAxisSize: MainAxisSize.max,
  mainAxisAlignment: MainAxisAlignment.spaceBetween,
  children: [
    Align(
      alignment: AlignmentDirectional(0, 0),
      child: Padding(
        padding: EdgeInsetsDirectional.fromSTEB(10, 0, 0, 0),
        child: Text(
          '${valueOrDefault<String>(
            (columnIndex + 1).toString(),
            '0',
          )}) ${valueOrDefault<String>(
            columnUserRow.username,
            'o',
          )}',
          style: FlutterFlowTheme.of(context).bodyMedium.override(
                fontFamily: 'Acme',
                color: valueOrDefault<Color>(
                  columnUserRow.id == currentUserUid
                      ? FlutterFlowTheme.of(context).primary
                      : FlutterFlowTheme.of(context).secondary,
                  FlutterFlowTheme.of(context).secondary,
                ),
                fontSize: 18,
                letterSpacing: 0,
              ),
        ),
      ),
    ),
    Padding(
      padding: EdgeInsetsDirectional.fromSTEB(0, 0, 10, 0),
      child: Text(
        valueOrDefault<String>(
          columnUserRow.points?.toString(),
          'o',
```



The snippet shows how the leardboard is implemented.

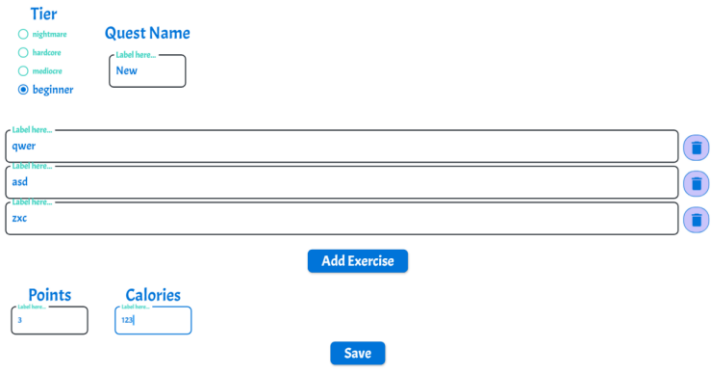
```
Padding(
  padding: EdgeInsetsDirectional.fromSTEB(0, 20, 0, 0),
  child: FFButtonWidget(
    onPressed: () async {
      await PostsTable().insert({
        'created_at': supaSerialize<DateTime>(getCurrentTimestamp),
        'title': _model.titleTextController.text,
        'description': _model.textController2.text,
        'userid': currentUserUid,
        'username': columnUserRow?.username,
        'porfile': columnUserRow?.profilePic,
      });
      context.goNamed('Feeds');
    },
    text: 'Post',
    options: FFButtonOptions(
      width: 200,
      height: 40,
      padding: EdgeInsetsDirectional.fromSTEB(24, 0, 24, 0),
      iconPadding: EdgeInsetsDirectional.fromSTEB(0, 0, 0, 0),
      color: FlutterFlowTheme.of(context).primary,
      textStyle: FlutterFlowTheme.of(context).titleSmall.override(
            fontFamily: 'Acme',
            color: Colors.white,
            fontSize: 20,
            letterSpacing: 0,
          ),
      elevation: 3,
      borderSide: BorderSide(
        color: Colors.transparent,
        width: 1,
      ),
      borderRadius: BorderRadius.circular(8),
    ),
  ),
)
```



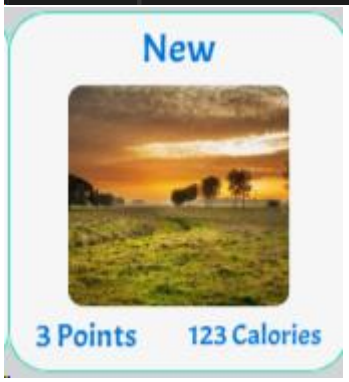
The snippet shows that when the post button is pressed, it save all the necessary data in the database which will then be show on the community page.

9.2 Admin Application

```
FFButtonWidget(
  onPressed: () async {
    await QuestsTable().insert({
      'exercise': _model.testModels.getValues(
        (m) => m.textController.text,
      ),
      'tier': _model.radioButtonValue,
      'name': _model.questNameTextController.text,
      'points_reward': int.tryParse(_model.pointsTextController.text),
      'calories': int.tryParse(_model.caloriesTextController.text),
    });
  },
  text: 'Save',
  options: FFButtonOptions(
    height: 40,
    padding: EdgeInsetsDirectional.fromSTEB(24, 0, 24, 0),
    iconPadding: EdgeInsetsDirectional.fromSTEB(0, 0, 0, 0),
    color: FlutterFlowTheme.of(context).primary,
    textStyle: FlutterFlowTheme.of(context).titleSmall.override(
          fontFamily: 'Acme',
          color: Colors.white,
          fontSize: 25,
          letterSpacing: 0,
        ),
    elevation: 3,
    borderSide: BorderSide(
      color: Colors.transparent,
      width: 1,
    ),
    borderRadius: BorderRadius.circular(8),
  ),
)
```

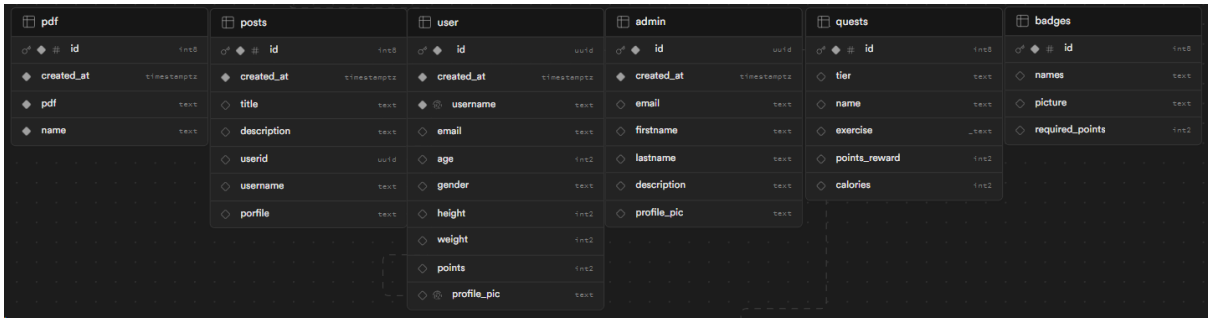


13	beginner	New	["qwer","asd","zxc"]	3	123



The snippet shows how a new quest might be created by the admin. When the quest is save into the database it will then be reflected back onto the user app and database.

9.3 Database



The database for FitQuest includes 6 tables. The user and admin table will store all relevant information about the user and admin. The pdf, quests and badges are independent tables which only stores information relevant to its table, this is then connect in the application itself. The post table has 2 foreign keys, meaning it is dependent on the user table.

10. Evaluation

Evaluation is an essential part of any application development process as it provides valuable insights into the usability, effectiveness, and overall user experience of the application. In the following section, the system is thoroughly evaluated using Nielsen's heuristics. These heuristics provide a set of general guidelines for evaluating user interface designs and were developed by usability specialist Jakob Nielsen.

To facilitate the review process, the Nielsen heuristics are organized in a tabular format. This allows for a clear and concise evaluation of each heuristic in relation to the functionality and design of the application. Based on how well the system adheres to Nielsen guidelines, each heuristic is evaluated independently. This review will uncover more information about the pros and cons of the system's user interface design, ultimately leading to improved user satisfaction and usability.

Visibility of System Status

Rating : 9

The system is clearly able to communicate the visibility of the system status. The system will automatically update the total points when completing a quest or the badges when sufficient points are earned. When navigating to different pages the system clearly indicates where the user is by the

title of the page or by indicating on the navigation bar at the bottom by highlighting the icon to identify the page one is on.

Match Between System and the Real World

Rating : 6

The design of the system uses both icons and images which the users is familiar with. However, the system also needs to also use words to help users if they don't identify the icon or image.

User Control and Freedom

Rating : 8

The application allows to freely navigate throughout the system. FitQuest does this by letting its users to start a quest whenever they want and not forcing them outside their time schedule. the system lets the user to choose how long they want their quest workout to last and edit their personal information whenever they want. The system successfully makes use of back buttons to make the navigation better.

Consistency and Standards

Rating : 7

The system maintains consistency to encourage intuitive actions which will reduce the cognitive load. Consistency is applied on the widget patterns, colour schemes (whether it is dark mode or light mode), icons on different pages. However, as this system's main focus is on pre-made workout routines as quest, it will be difficult for users to learn the functionalities of this app.

Error Prevention

Rating : 5

Error messages are incorporated into the design but to the very minimal. The system needs major work on error prevention methods. There are error prevention methods placed on the login and register page in both the user and admin system.

Recognition Rather than Recall

Rating : 8

The system successfully minimizes the memory load by making actions and information simple and intuitive.

Flexibility and Efficiency of Use

Rating : 10

The system accommodates to different user preferences that provides a flexible user experience. The system also provides dark mode and light mode which help users to choose their preference according to their setting or choice.

Aesthetic and Minimalistic Design

Rating : 10

The interface design developed for FitQuest is clean and aesthetically pleasing, which does not divert the user's attention away from the main goal of the system. The system follows a simple colour scheme, fonts and a presentable layout which grants the system a minimalistic and clean design.

Helps Users Recognise, Diagnose, and Recover from Errors

Rating : 4

The FitQuest design and user interface barely provides users with informative error messages that will address any potential error made by the user. This is a major issue as it will cause the user to be confused about why they are facing an error.

Help and Documentation

Rating : 8

As the system can get confusing at times. FitQuest will provide new users with guides to understand the features and functionalities of how to operate the system. This is provided in setting where the user can find help documents or contact the administrator of the system,

In the Heuristics the system scored a **75**, which is pretty good for developing a prototype system. The system will require additional work and future development, to make the overall usability of the system better.

11. Critical Appraisal

11.1 Process

During the entire development of this project, a plan was developed to meet all the deadlines for each project section. Although the overall progress was smooth, challenges arose regarding time management and research. The part that increased the duration past the project section deadline was the research and justifying how each section of the research is important for the development of the product.

The initial stages of the project required choosing the methodology and framework for this project. The selected methodology and framework were Agile with DSDM, which was the best methodology considering the time constraints faced when developing a product in quick successions. This helped by breaking the project into sizeable chunks, allowing to go back to the previous stage if something in the previous stage was incorrect.

The literature review and product research heavily impacted the development the product. The literature review helped identify what was the problem statement and how the to solve that problem. It helped by understanding what individuals required and the product research further added to understanding how the requirements can possibly be implemented. The design process was made easy due to the research done in the pervious sections.

The hardest part of the project was implementing the entire system. This is because it is easier to visualize an application, but when it comes to implementing it, it becomes very difficult. Overall, the entire project development went smoothly, even with all the hiccups along the way.

11.2 Product

The aim of the project was to develop a system that adopts gamification features to address the lack of motivation in individuals who often encounter when trying to engage in physical activities. Even

though the system checks all the requirements functionalities, however, it does not mean that the system is perfect. After evaluating with the Nielsen's heuristics, it can be noticed that there are still many flows in the system which need to be worked on. Furthermore, the system still needs to be worked on and needs to have future development.

11.3 Personal

My personal views when developing this product would be that when making a similar project in the industry would not be as easy. This is because, there will be extensive research, reviews, interviews and testing which make the system to have many time constraints. I built this system because it reflected my own standing in real life. There have been multiple occasions where I wanted to start working out but have been unable to do so due to lack of motivation. Furthermore, I wanted to create a system that does focus not on the pedometer as the main focus and rather the stationary exercise.

Furthermore, I have no experience in building an app and only have experience building website, and am very weak in writing codes, that is the reasoning I have used Flutterflow which is a code generator where I only had to drag the widgets and manually configure to the back end (Supabase).

This project did teach me new research techniques and various parts in software development. I have noticed a personal and academical growth during the development of this project which helped me understand the roles of a researchers, analyst, designed and developer.

11.4 Future development

- The system still needs major development, as there are still many holes in the existing system. The holes were explained in the technical project development and will first need to be solved and patched.
- The user application needs to allow users to customize their own workout, as pre-made workout would prove to be quite unsatisfactory to some users. However, a major problem to that would be is how will the points be allocated for it. This can be solved by having the customizable workout have the same points value.
- In the community page, the implementation of likes and comments would be excellent as it would allow users to interact with each other.
- To further develop the capabilities of this system, a machine learning model can be placed, where it can learn the user’s action and activities to create a custom work suiting their preference. This can cause a problem with the Data Protection Act and will require to be stated in the policies when the user signs up.
- Integrating a pedometer and calorie tracker to the app would be allow the app to be an all-in-one app. This will make it so that it has all the functionalities that the researched fitness app had.
- Allowing the user to integrate and sync their own personal fitness wearable devices to the app would be a great addition. As it would allow the users to stay comfortable and not exercise with their mobile device.
- The admin system can be further developed by adding functions like editing existing workout quest or deleting inappropriate post made by the users.

12. Conclusion

Overall, the system aims to deliver a platform that not only motivates and facilitates physical activities but also does it within the legal requirements, cultural norms, ethical standards, and best industrial practices, by maintaining these considerations, this system can guarantee a responsible, inclusive, and professional use to promote a better physical well-being of its users while also contributing to a healthier and active society.

The initial investigation unfolded the vastness of the health and fitness industry and the IT development industry. This helped to understand that both these industries are constantly changing, and it is very important to stay well informed about them.

To conclude the project's main objective was to develop a fitness app that leverages gamification to motivate users to start working out or continue to work out, which would promote a healthier lifestyle. As the product research revealed that there are many apps that promote gamified fitness which involves moving around more and not stationary workouts, this prompted an idea to develop an app that helps user with stationary workouts. Even though the app cannot be compared to industry standards and requiring multiple developments to reach that standard as explain in the above sections. The app does successfully fulfil the main aims of the objective.

References

Capel, T., Schnittert, J. F., Snow, S. and Vyas, D. (2015) Exploring motivations of young adults to participate in physical activities, In *Conference on Human Factors in Computing Systems - Proceedings*, Association for Computing Machinery, pp. 1409–1414.

Düsenflugzeug, J. (2022) You Say You Want A Resolution, *Post-Its*, p. 24.

Feng, W., Tu, R. and Hsieh, P. (2020) Can gamification increases consumers' engagement in fitness apps? The moderating role of commensurability of the game elements, *Journal of Retailing and Consumer Services*, Elsevier Ltd, 57.

Huang, X. (2022) Data Visualization Design Strategies for Promoting Exercise Motivation in Self-Tracking Applications, In *Proceedings of the 40th ACM International Conference on Design of Communication, SIGDOC 2022*, Association for Computing Machinery, Inc, pp. 78–89.

Leung, K. K., Sick, K., Huellemann, K. L. and Pila, E. (2023) Body image flexibility and exercise motivation: A two-sample replication study, *Body Image*, Elsevier Ltd, 46, pp. 212–222.

Looyestyn, J., Kernot, J., Boshoff, K., Ryan, J., Edney, S. and Maher, C. (2017) Does gamification increase engagement with online programs? A systematic review, *PLoS ONE*, Public Library of Science, 12(3).

McGonigal, K. (2011) How Can Clients Keep Their New Year's Resolutions?, *IDEA Health & Fitness Association*, [online] Available at: https://www.ideafit.com/personal-training/how-can-clients-keep-their-new-yearrsquos-resolutions-2/ (Accessed 15 January 2024).

Neupane, A., Hansen, D., Sharma, A., Fails, J. A., Neupane, B. and Beutler, J. (2020a) A Review of Gamified Fitness Tracker Apps and Future Directions, In *CHI PLAY 2020 - Proceedings of the Annual Symposium on Computer-Human Interaction in Play*, Association for Computing Machinery, Inc, pp. 522–533.

Neupane, A., Hansen, D., Sharma, A., Fails, J. A., Neupane, B. and Beutler, J. (2020b) A Review of Gamified Fitness Tracker Apps and Future Directions, In *CHI PLAY 2020 - Proceedings of the Annual Symposium on Computer-Human Interaction in Play*, Association for Computing Machinery, Inc, pp. 522–533.

Ozdamli, F. and Milrich, F. (2023) Positive and Negative Impacts of Gamification on the Fitness Industry, *European Journal of Investigation in Health, Psychology and Education*, Multidisciplinary Digital Publishing Institute (MDPI), pp. 1411–1422.

Rajšp, A., Heričko, M. and Fister, I. (2020) *The Use of Gamification in Smart Sport Training*,.

Razali, N., Nasir, N. A., Ismail, M. E., Sari, N. M. and Salleh, K. M. (2020) Gamification Elements in Quizizz Applications: Evaluating the Impact on Intrinsic and Extrinsic Student's Motivation, In *IOP Conference Series: Materials Science and Engineering*, IOP Publishing Ltd.

Sebastian, S., Florian Supervisor, B. and Klaus, O. (2021) *Rewards in Gamification: A Conceptual Replication and Extension*,.

Speakman, J. R. (2020) An Evolutionary Perspective on Sedentary Behavior, *BioEssays*, John Wiley and Sons Inc., 42(1).

Sun, M., Jiang, L. C. and Huang, G. (2023) Improving Body Satisfaction Through Fitness App Use: Explicating the Role of Social Comparison, Social Network Size, and Gender, *Health Communication*, Routledge, 38(10), pp. 2087–2098.

Wang, J. Q., Cong, J. chen, Zheng, Z. yuan, Meng, Y. and Liu, C. (2019) An Interaction Design Approach of Fitness APP, In *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*, Springer Verlag, pp. 348–358.

Webley, K. (2012) Lose Weight and Get Fit - Top 10 Commonly Broken New Year's Resolutions - TIME, *Time*, [online] Available at: https://content.time.com/time/specials/packages/article/0,28804,2040218_2040220_2040221,00.html (Accessed 15 January 2024).

Wei, C. (2017) *Integrating gamification to modern web fitness services*,.

Yamani, H. A. (2021) A Conceptual Framework for Integrating Gamification in eLearning Systems Based on Instructional Design Model, *International Journal of Emerging Technologies in Learning*, Kassel University Press GmbH, 16(4), pp. 14–33.

Zhou, M., Mintz, Y., Fukuoka, Y., Goldberg, K., Flowers, E., Kaminsky, P., Castillejo, A. and Aswani, A. (2018) *Personalizing Mobile Fitness Apps using Reinforcement Learning HHS Public Access*,.

Appendix 1: Project Proposal

1. Overview

There is a frequent occurrence in the initial eagerness to start working out, which is usually sparked by one's individual goals or New Year's Resolutions, which rapidly dims due to a lack of motivation. According to (Webley, 2012), one of the most commonly broken New Year Resolution is “Loss Weight and Get Fit” which is further elaborated by (McGonigal, 2011) that a large number of people would sign up to gyms in January, but that number would be diminished by mid-February.

There are many different motivational factors that affect people performing any physical activities. (Capel et al., 2015) asserted that there are primarily two types of motivations for engaging in physical activity. The first group includes things like tradition in the family, competitive motivation, parental support, socialising, goal-setting, tracking, and athletic background that are related to initial sports participation. The second group of factors, which included satisfaction of sports engagement, a sense of wellbeing, self-competition, interacting with others, health, and an abiding appreciation of nature, led to continuous sports activity.

Both (McGonigal, 2011) and (Düsenflugzeug, 2022) describes that people who get into fitness have set unrealistic goal which often clashes with their intention to indulge in physical activity. People require a combination of encouragement and realism which gives them the best chance of reaching their goals. This shows that many people run into multiple difficulties while trying to start a regular fitness schedule.

(Ozdamli and Milrich, 2023) explains that gamification features help motivate individuals to work out and that it has become a trend in the fitness sector. It is further emphasized that these gamification features add fun and a competitive element to regular workout routines which inspire and motivate people to achieve their fitness goals.

(Neupane et al., 2020) states that many activity trackers have been using gamification features. It is agreed by both (Ozdamli and Milrich, 2023) and (Neupane et al., 2020) that gamification features significantly help in motivating people to improve their health.

A potential solution to this issue would be to develop a system that aims to increase and maintain motivation, get past common obstacles to motivation (such as responsibilities, monotony, and detachment from fitness goals), improve user engagement through challenges and achievements, encourage a healthier lifestyle, and promote knowledge of the role that gamification plays in encouraging people to exercise and, ultimately, improve their health and well-being.

Keywords: [Fitness, Motivation, User-Interface, Feedback, Gamification, Game, System-Development]

2. Aim

This project aims to develop a system that adopts gamification features to address the lack of motivation in individuals who often encounter when trying to engage in physical activities.

3. Objectives

3.1 Research and Information Gathering [23 Days]

3.1.1 Literature Review [16 Days]

3.1.1.1 Research how motivation plays an important role when working out. [2 Days]

3.1.1.2 Research how fitness trackers play an important motivational role in the fitness industry. [3 Days]

3.1.1.3 Research how gamification features affect motivation when it comes to physical activities. [2 Days]

3.1.1.4 Explore how gamification features are being added to fitness trackers. [2 Days]

3.1.1.5 Explore Human Computer Interaction(HCI) with gamification. [2 Days]

3.1.1.6 Explore different visualisation techniques which can motivate individuals within the fitness systems. [2 Days]

3.1.1.7 Research different types of policies. [3 Days]

3.1.2 Product Research [7 Days]

3.1.2.1 Research pre-existing gamified fitness systems and their functionalities using Nielsen's Usability Evaluation. [3 Days]

3.1.2.2 Research different technologies to build the system. [2 Days]

3.1.2.3 Document system interaction to understand usability functions for a testing plan. [2 Days]

3.2 Design Documentation [21 Days]

3.2.1 Identify the functional and non-functional requirements. [2 Days]

3.2.2 Requirement Prioritization using MoSCoW. [3 Days]

3.2.3 Create a rich picture diagram to identify all relating factors involved with system. [2 Days]

3.2.4 Create a Use-Case diagram to identify how actors will use the functionalities of the system. [3 Days]

3.2.5 Create a Conceptual ERD to identify entities, attributes, and relationships with the system. [2 Days]

3.2.6 Create a Physical ERD to define columns and keys within each entity to map out the data model. [3 Days]

3.2.7 Create a database schema to visualize the entire relational database. [3 Days]

3.2.8 Create a sequence diagram to understand possible user interaction. [3 Days]

3.3 Iterative Prototype Development [17 Days]

3.3.1 Documentation of required HCI elements. [2 Days]

3.3.2 Build wireframes to understand the layout of the system. [3 Days]

3.3.3 Create a low-fidelity prototype. [3 Days]

3.3.4 Modify prototype using feedback from actors. [2 Days]

3.3.5 Create a mid-fidelity prototype. [5 Days]

3.3.6 Test mid-fidelity prototype and correct any possible errors. [2 Days]

3.4 Product Implementation and Development [32 Days]

3.4.1 Implement the database schema. [5 Days]

3.4.2 Implement the back-end. [10 Days]

3.4.3 Implement the front-end by creating a system for users and administrators. [14 Days]

3.4.4 Develop required policies and procedures. [3 Days]

3.5 Testing [11 Days]

3.5.1 Create a usability testing plan. [2 Days]

3.5.2 Test the system based on the plan. [2 Days]

3.5.3 Document database, white-box, and black-box testing. [5 Days]

3.5.4 Document possible errors and how they will be corrected. [2 Days]

3.6 Evaluation Report and Future Implementation[8 Days]

3.6.1 Evaluate which functionalities were accomplished when developing this system. [2 Days]

3.6.2 Evaluate the product based on the Nielsen's Usability Evaluation. [2 Days]

3.6.3 Evaluate the limitations of the system and how to improve these drawbacks. [2 Days]

3.6.4 Evaluate any future implementations. [2 Days]

3.7 Project Framework or Any Methodology used

The project will adopt the agile methodology which is more flexible and adaptable throughout the development process. Regular user feedback and iterative changes will also be possible with this technique. This methodology will guarantee that adding gamification elements to a fitness-based system will be done in a systematic, user-centred, and flexible manner.

4. Legal, Social, Ethical and Professional

4.1 Legal

When the system is being developed, there will be several legal considerations that will be needed to be addressed in order to guarantee that the system follows all applicable laws and regulations, by complying with the Data Protection Act 2018. This project will safeguard user data by obtaining adequate authorization for data processing and providing transparency to users about how their data is collected, stored, and processed within the system.

Adhering to the General Data Protection Regulation(GDPR) is of the utmost importance. This will provide the basic framework for the developers to uphold user rights, which includes the right to access data, request data, delete data, and share data with other parties. All these legal frameworks will enable the system to operate within the boundaries of the law ensuring the users' data and rights are safeguarded while offering a secure platform for their fitness regimen.

4.2 Social

During the development of this system, the developers will need to consider a number of social factors in order to build a platform that can appeals with the users and promote a sense of inclusivity and community. Cultural sensitivity will be an essential consideration when developing this system. Considering the diverse cultural backgrounds and perspectives of its users, the system will need appropriate engagement strategies to maximize user interaction.

Furthermore, the system will actively encourage community engagement which aligns with the public health campaigns. This will help enhance a sense of community and social support, whereby the system improves overall user experience and motivates individuals to participate in physical activities. This will create a platform that dynamically engages with the local communities. This approach will enhance user experience and provide value to the diverse user background.

4.3 Ethical

Ethical considerations will be vital when building this system. Some of the areas of considerations will focus on user data privacy and the influence of gamification elements. The core ethical structure is transparency and respecting users' data rights. This means that the developers will need to prioritize transparency on how data is handled. Respecting users' rights is not only a legal requirement but an ethical obligation.

The developers will have to be cautious when implementing gamification elements to avoid unintentionally encouraging excessive exercise or unhealthy habits. It will be important to find the ideal balance to encourage healthy and sustainable fitness behaviours rather than driving users to extremes. By prioritizing these ethical moralities, the system will offer a supportive and well-rounded platform for users to achieve their fitness goals.

4.4 Professional

The developers will need to follow to professional standards and practices. Cybersecurity and quality assurance will serve as the pillars of professionalism throughout the development of this system. This includes rigorous quality assurance processes, conducting security audits and regular penetration testing to guarantee user data is protected and to assert user trust.

Regulatory updates and compliance will be a fundamental part of professionalism. Where the developers must stay up to date on the ever-changing regulations, concerning data protection and privacy. Adapting the system to comply with the latest legal requirements is not only a matter of professionalism but also a legal obligation. By upholding these professional considerations, the system can provide a high-quality and reliable platform for users while also in compliance with the best industrial practices.

4.5 Conclusion

Overall, the system aims to deliver a platform that not only motivates and facilitates physical activities but also does it within the legal requirements, cultural norms, ethical standards, and best industrial practices, by maintaining these considerations, this system can guarantee a responsible, inclusive, and professional use to promote a better physical well-being of its users while also contributing to a healthier and active society.

5. Planning (see appendix A)

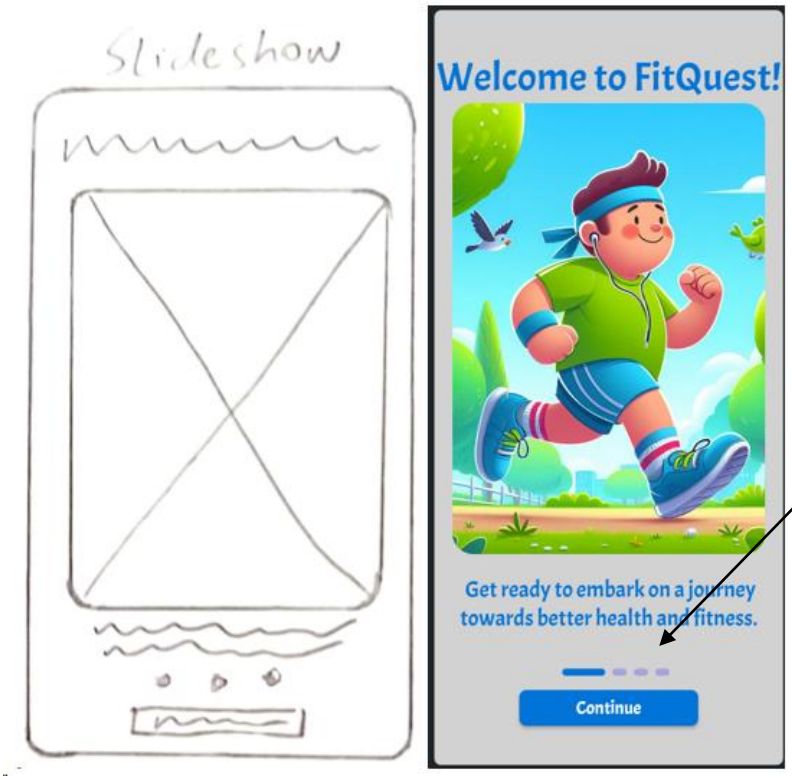
In order to build a flexible and dynamic framework, the project will use the Agile Rooted Methodology in conjunction with the iterative development approach. The agile methodology will allow frequent feedback and adjustable changes, which make the development more responsive to both user needs. During the development of this system, the developers will be using evolutionary

prototyping, which will first start with an initial basic prototype, which will further evolve and expanded upon the user feedback and requirements.

The developers will utilize Test-Driven Development(TDD) to prioritize more on quality and reliability which will concentrate on the quality of the code. Furthermore, the development team will apply UML (Unified Modelling Language) to design the system structure to foster efficient and effective development.

The development team will follow an iterative development process. This method allows for more flexibility and adaptation which associates the system closely with changing user requirements. By following this comprehensive approach, the system seeks to provide a high-quality and responsive fitness system to address the user needs effectively.

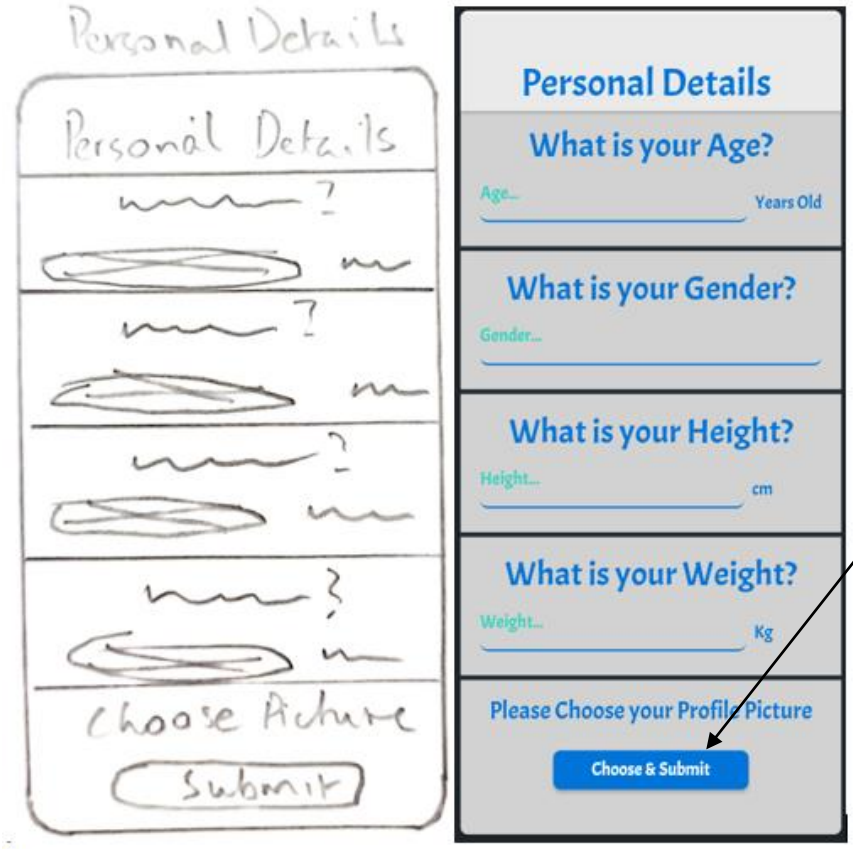
Appendix 2 Comparison of Wireframes, Mid-Fidelity and Final Product



This image shows when a new user uses this app, they are first greeted, and explained on what to expect from this app. This is presented in a slideshow format.



After the user is through with the slideshow, the user will either have to create an account or login



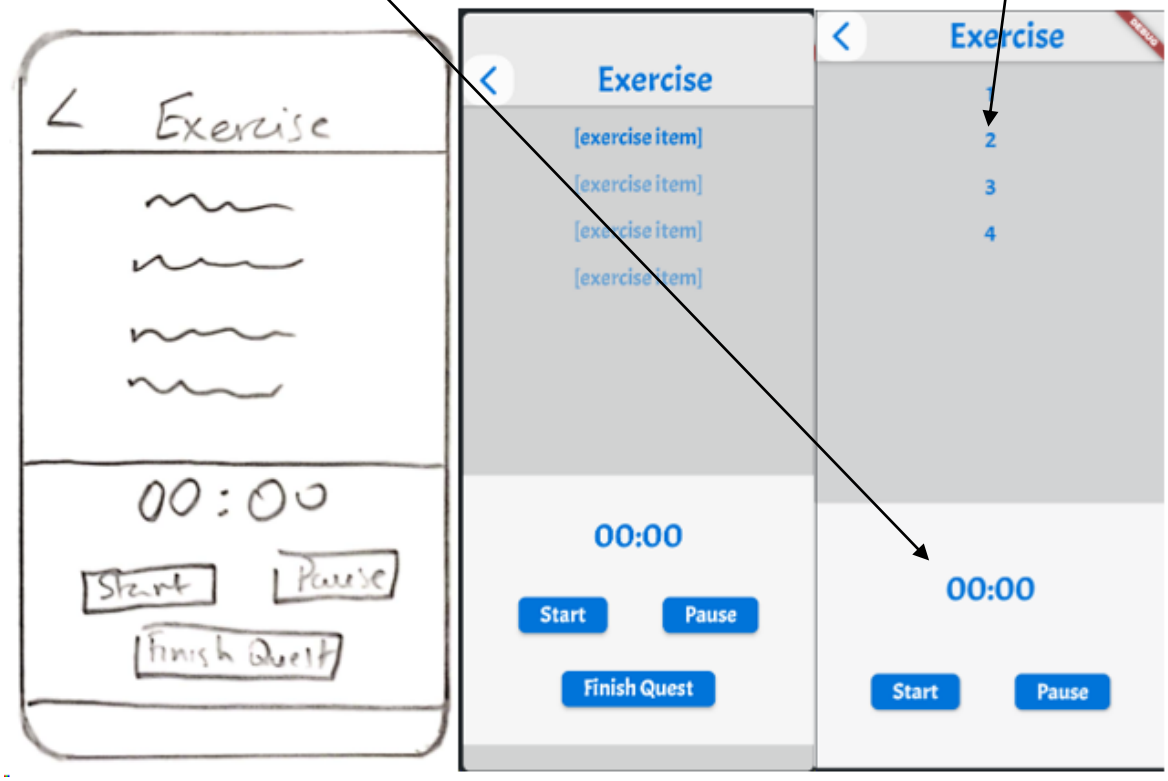
If the user creates an account, they have to enter their details and submit a picture.

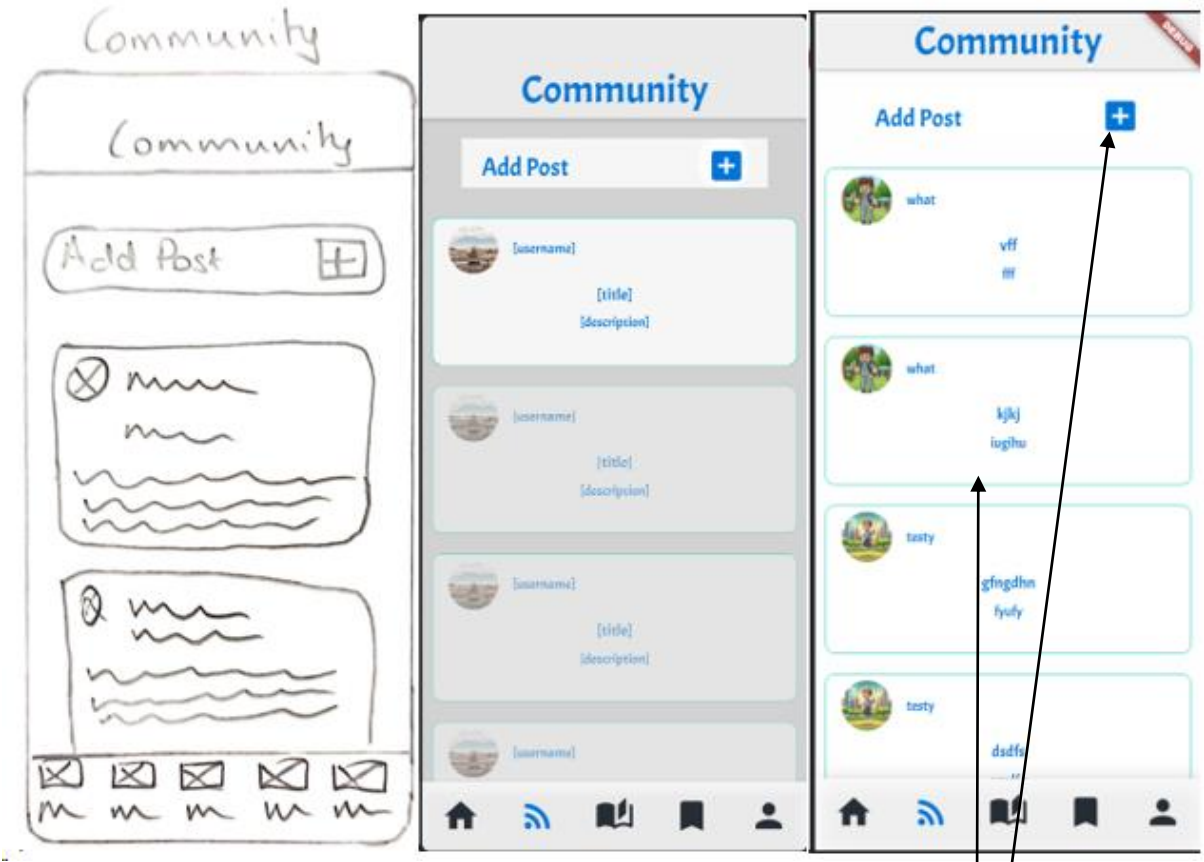


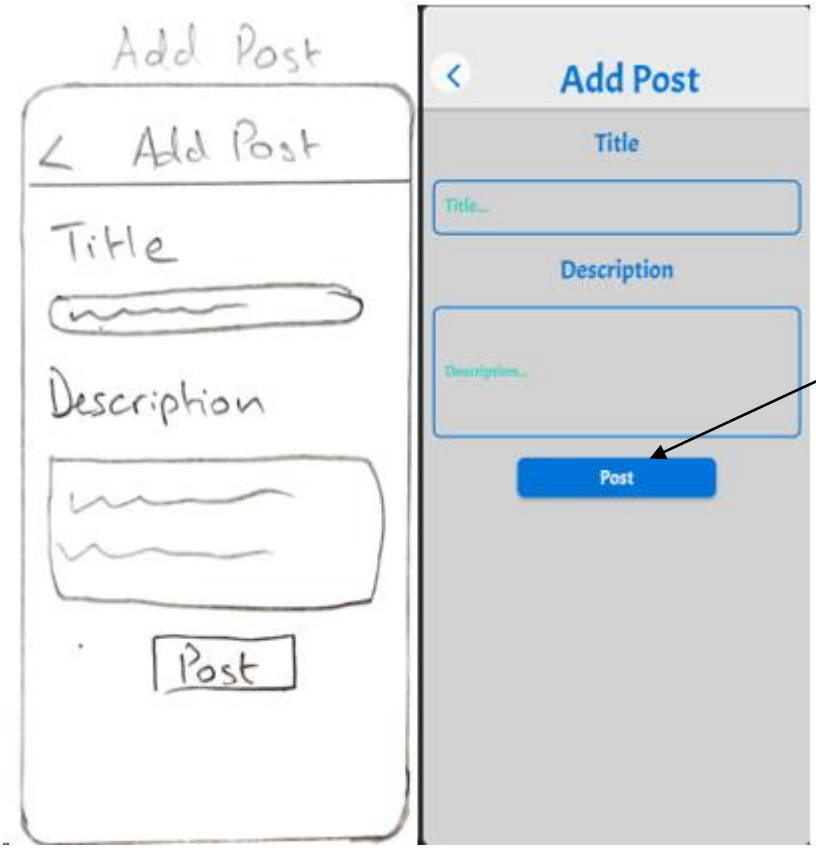
On this page the user can choose the difficulty of the quest that they want to start. After choosing one of the difficulties, they will be navigated to the choose difficulty's quests.



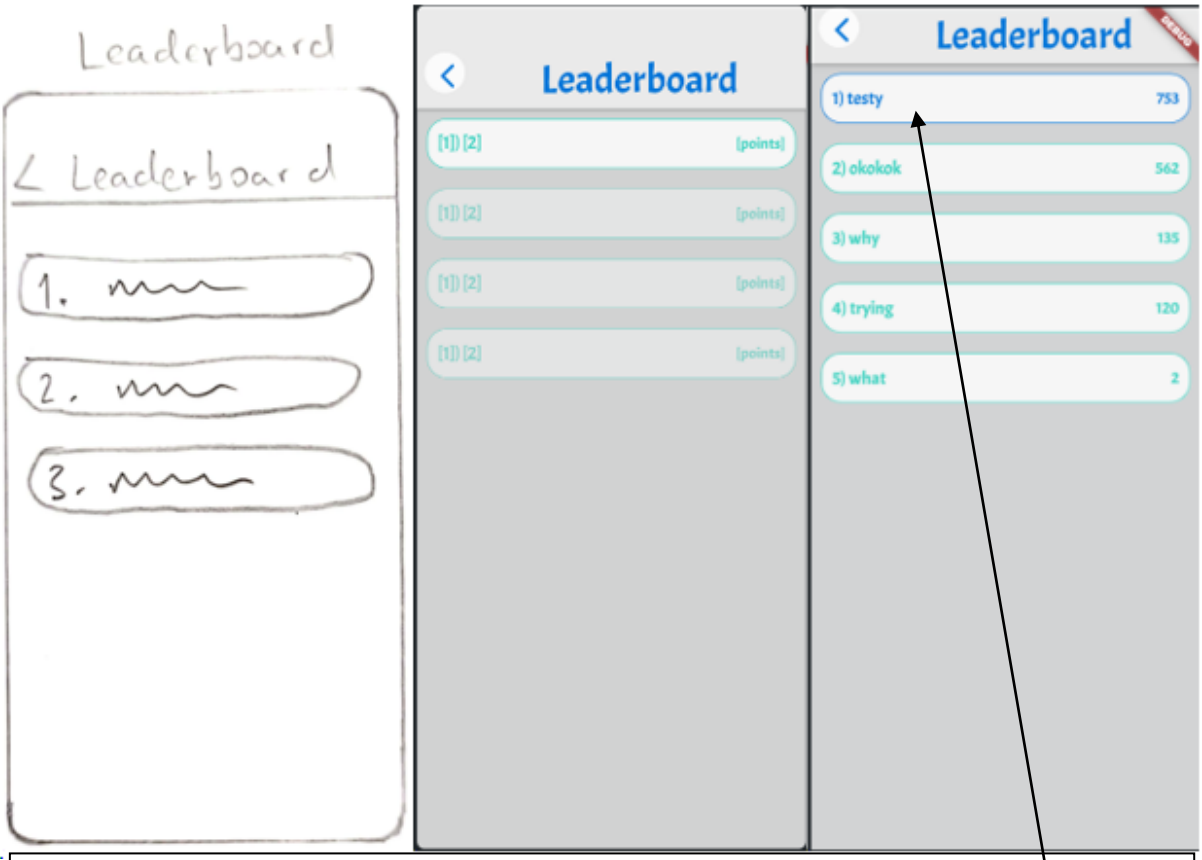
This image shows the progression of the application from it's wireframes to the final product. After choosing the difficulty, the user will be prompted to choose a quest from that difficulty and start the quest. After starting the quest, the user can view the exercises while also starting, pausing and ending the timer.







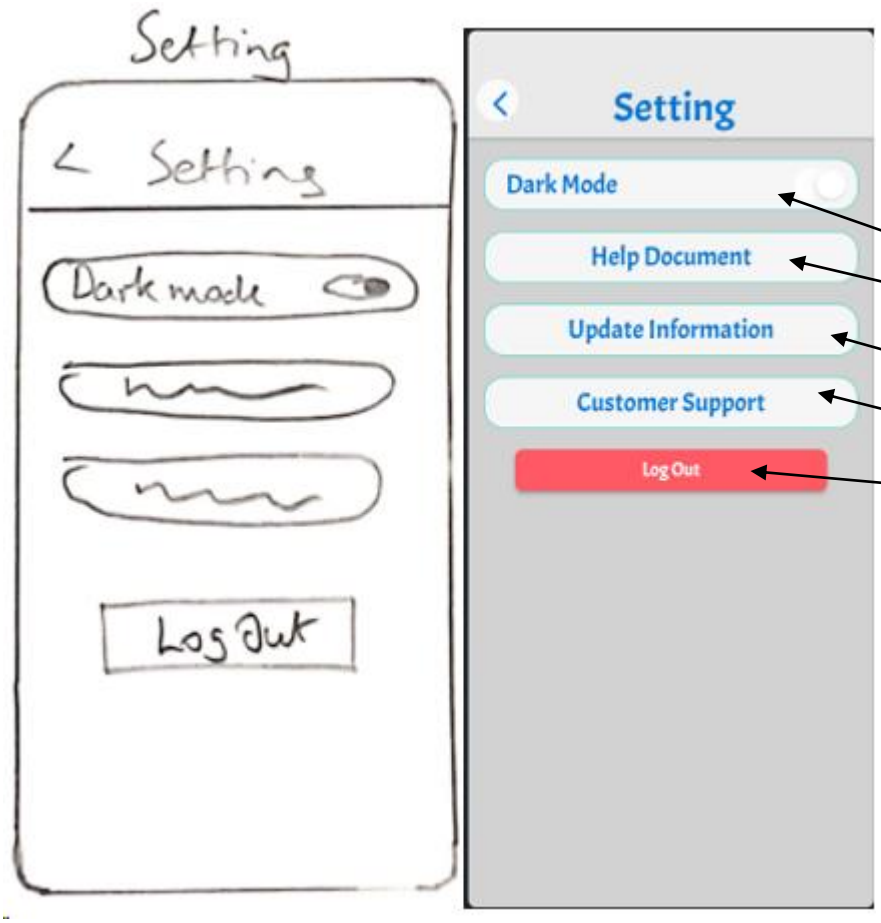
The user on this page can view all the post made here. Furthermore, they choose to also add a post, by pressing the Add Post button and posting it



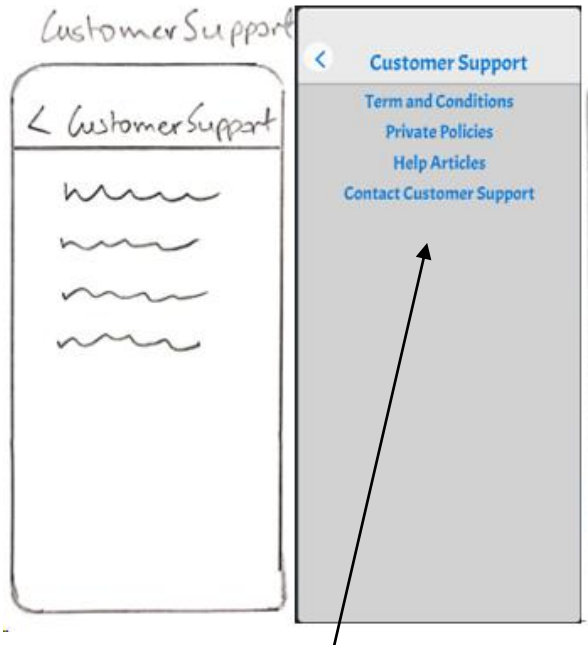
The user can view the rankings of all the players and also see their own ranking, which has a different colour.



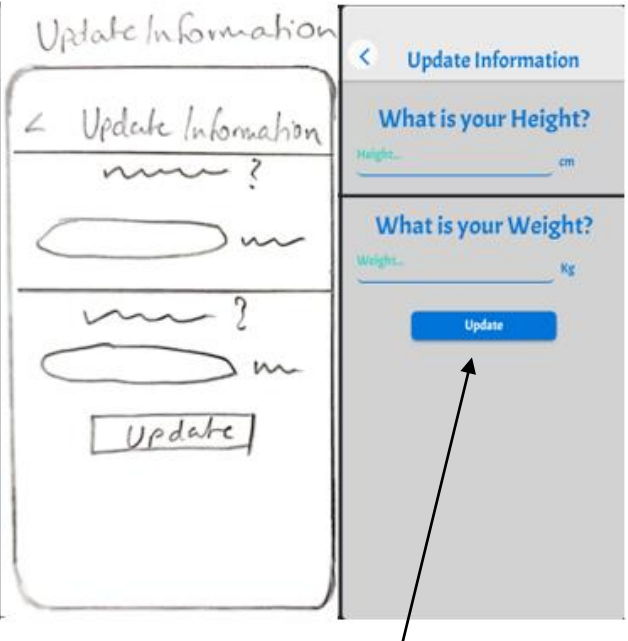
On this page the user can view their personal details as well as update their profile picture and navigate to the settings.



In the setting the user can go into dark mode, view help documents, update their details, call customer support or log out.



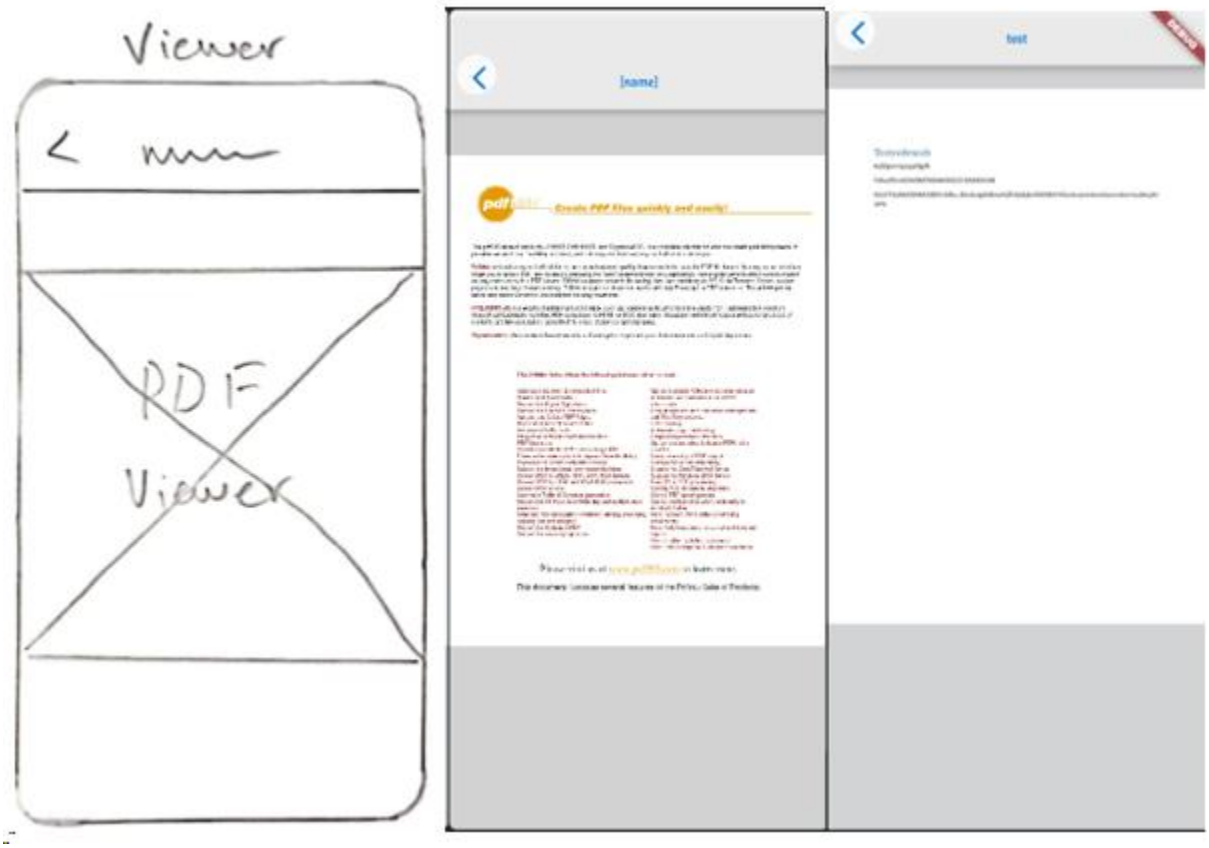
The customer support page.

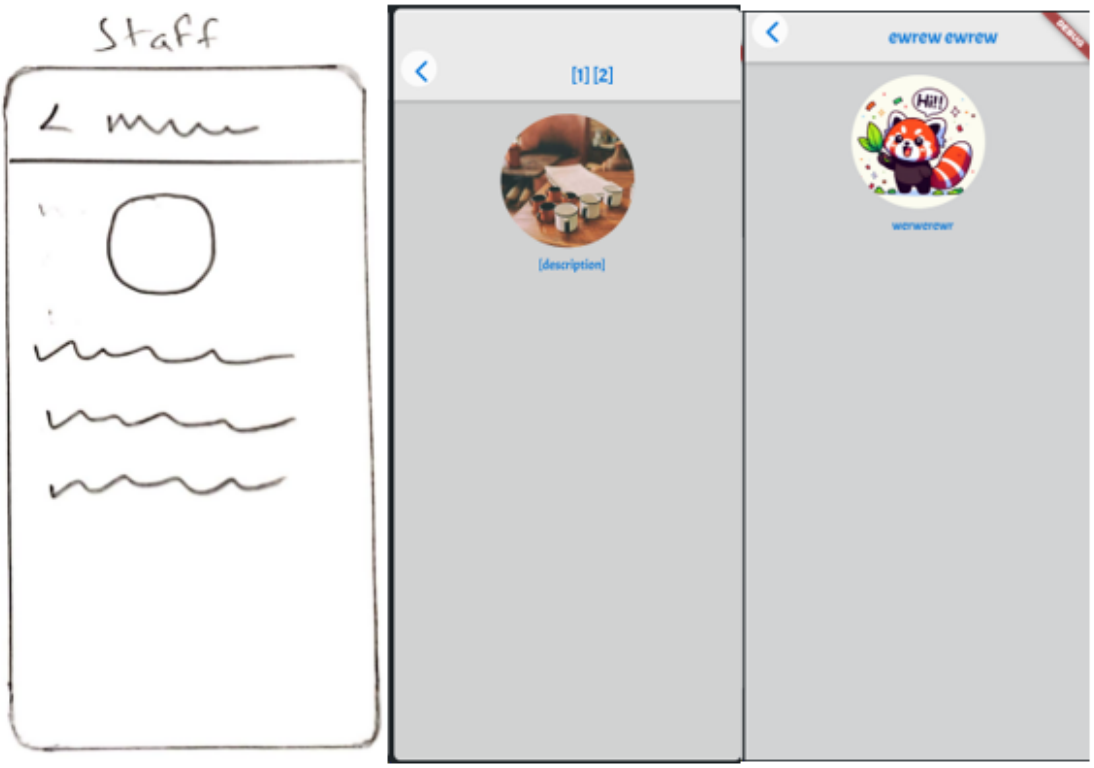


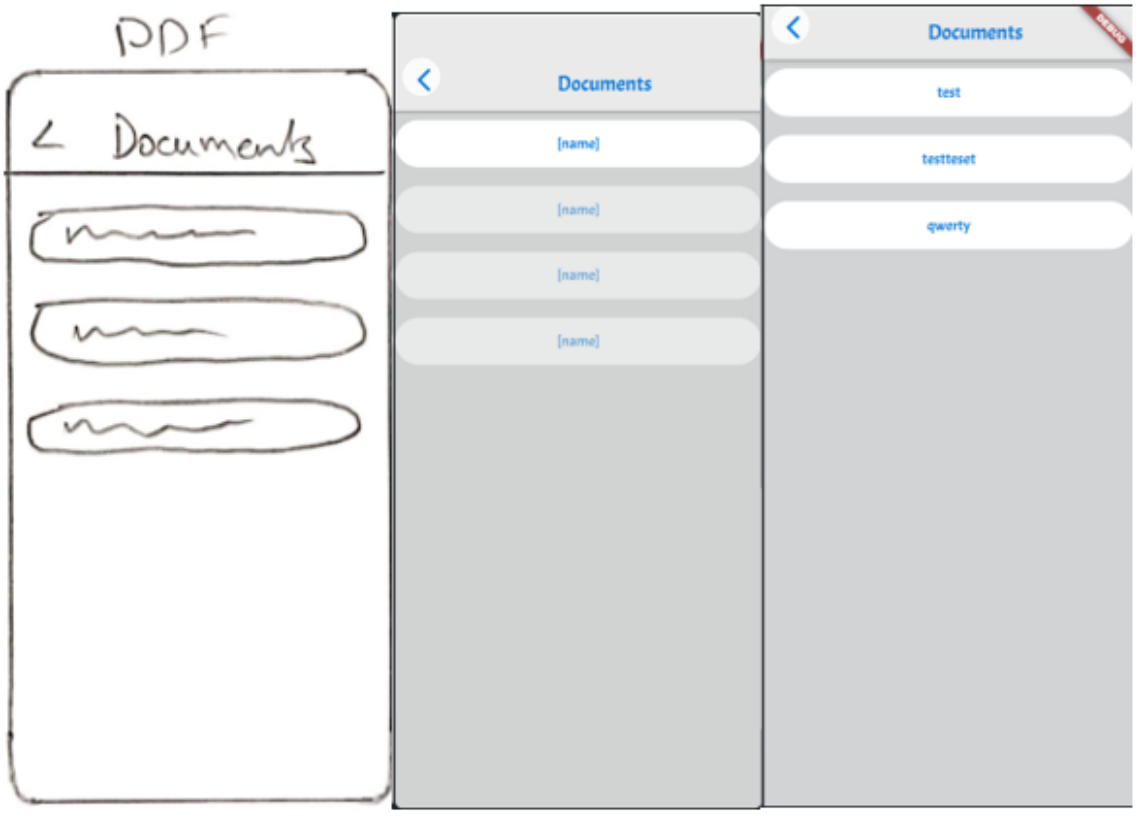
Update personal information page.

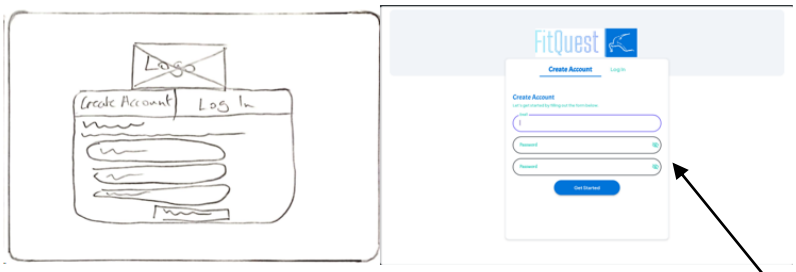


On the admin page the user can view different admins and selected them and view their description, while they are also able to select the documents button and select a document to view them.



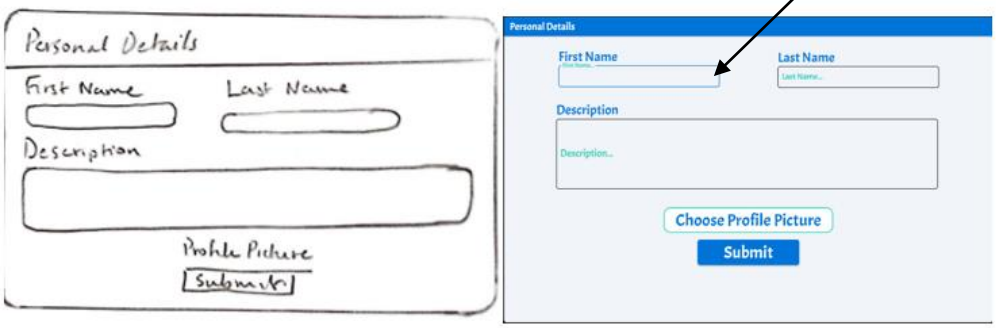

Staff
[1] [2]
[description]
ewrew ewrew
Hi!!

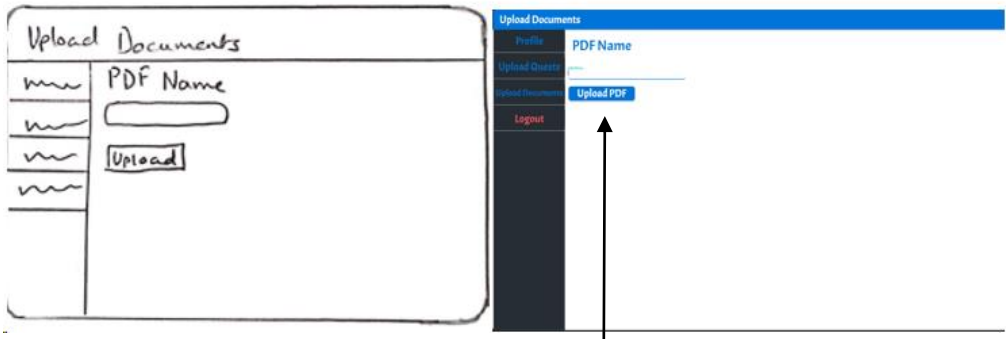

PDF
Documents
Documents
[name]
[name]
[name]
[name]
Documents
test
testteset
qwerty



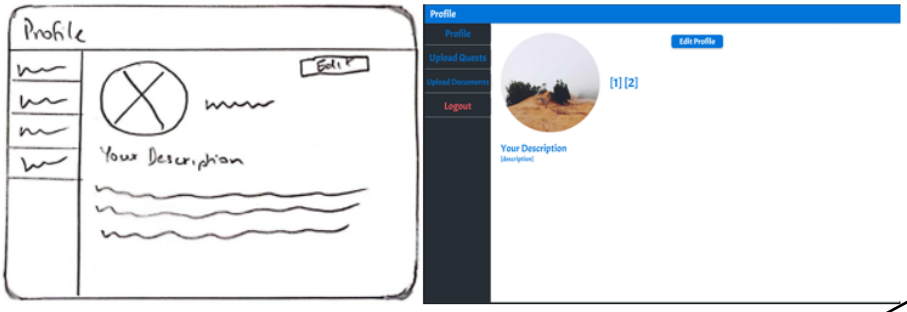


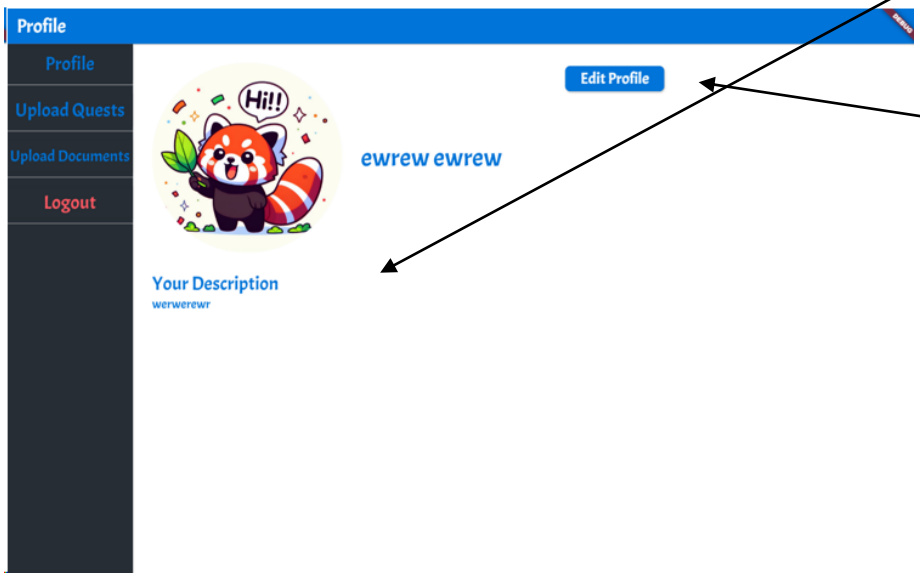
The admin has a similar interface for when the admin wants to register a new account or login. Similarly there is also a personal details page.





The admin can upload pdf documents on this page





This is the profile page which show the admins details, this page also has the edit profile button



The page allows the admin to upload quest into the system